GENETIC TECHNIQUES & ITS USES IN MEDICINE

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Objectives:

At the end of this chapter, it is expected;

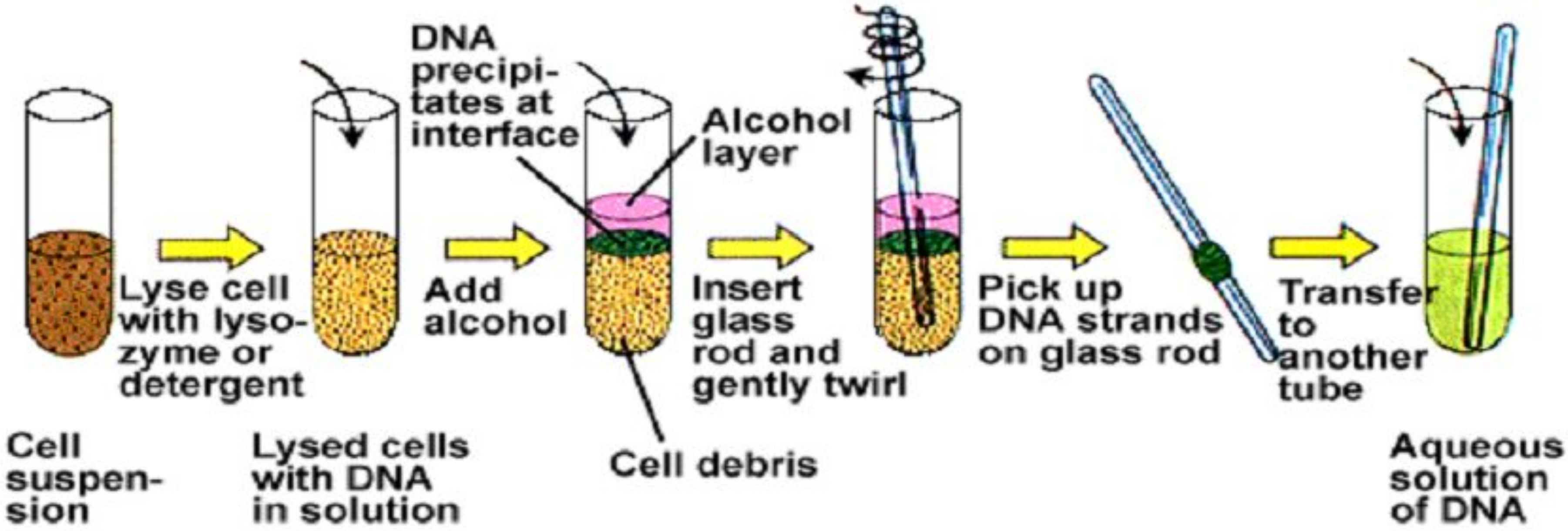
To describe the methods and applications of the following genetic engineering techniques:

- Restriction Enzymes
- DNA probes
- Blot transfer techniques.
- Polymerase chain reaction (PCR),
- Restriction fragment length polymorphism and its applications.
- Fluorecent in-situ hybridization (FISH).
- DNA sequencing.
- DNA fingerprinting and DNA foot-printing.

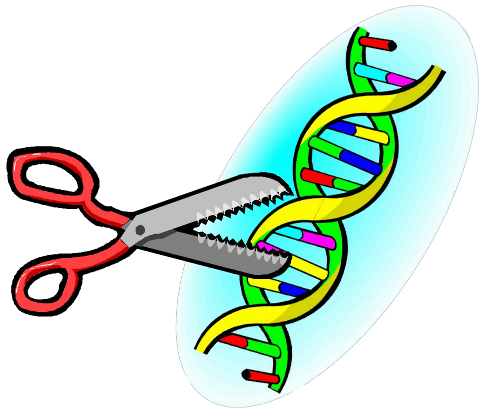
Manipulating DNA

- Techniques used to manipulate DNA:
  - DNA Extraction
  - Cut DNA in to smaller pieces
  - Identify base sequences
  - Make unlimited copies of DNA

DNA Extraction



Cut DNA in to smaller pieces
- Restriction enzymes

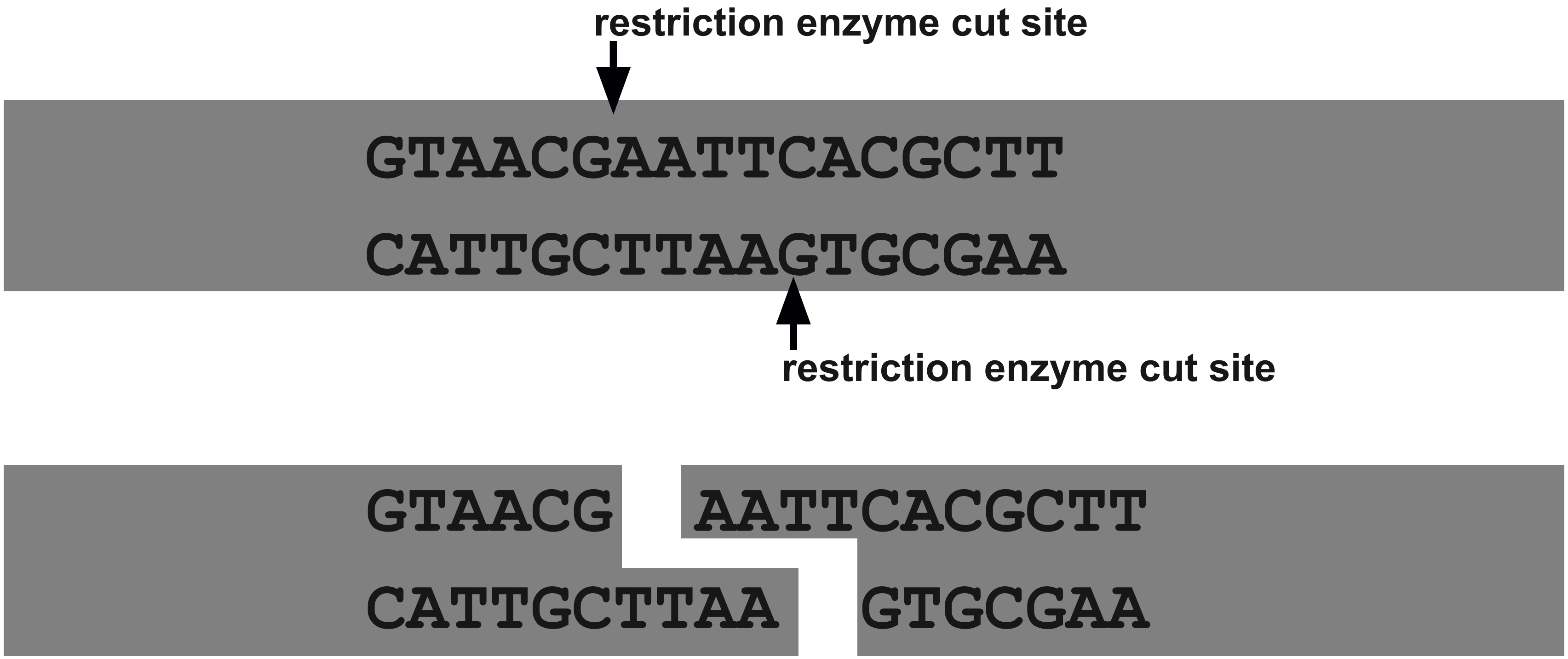


- DNA "scissors"
  - Used by bacteria to chop up DNA of attacking viruses
  - Eg., EcoRI, HindIII, BamHI
- – cut DNA at specific sites
  - Enzymes look for specific base sequences

GTAACG | AATTCACGCTT
CATTGCTTAA | GTGCGAA

Restriction enzymes

- Cut DNA at specific sites



Characteristics of Restriction Enzymes

1. Cut DNA sequence-specifically

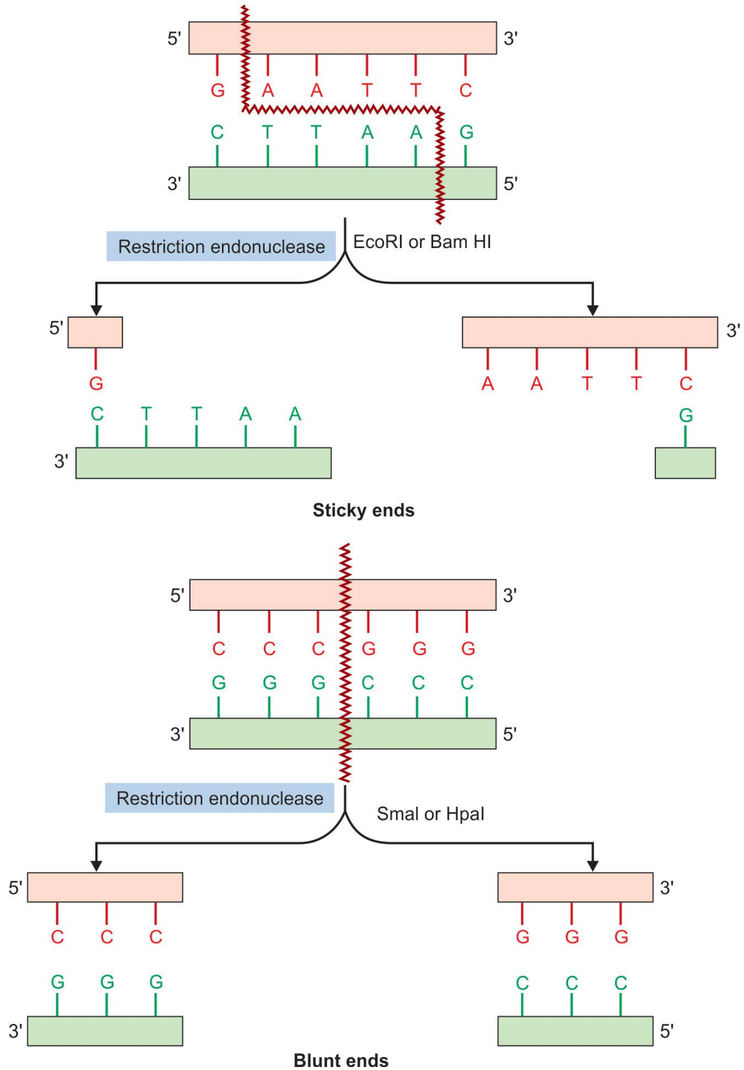
2. Bacterial enzymes; hundreds are purified and available commercially

3. Restriction-modification system Bacteria have enzymes that will cleave foreign DNA; hence, "restrict" the entry of viral DNA. To prevent the bacteria's own DNA from being cut, there is a second enzyme that methylates the same sites recognized by the restriction enzyme (modifies that site).

4. Named (e.g., *Eco*RI) for bacterial genus, species, strain, and type

5. Recognize specific 4-8 bp sequences
    - sequences have symmetry (they are palindromes)
    - after cutting the DNA, the cut ends are either
        - blunt
        - staggered (overhangs) - cohesive ends facilitate cloning the DNA

Action of restriction endonuclease



- EcoRI or BamHI produces sticky ends
- SmaI or HpaI produces blunt ends

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Types of Restriction Enzymes

- Type I enzymes (EC 3.1.21.3):
  - Cleave at sites remote from recognition site;
  - e.g. EcoK, EcoB
- Type II enzymes (EC 3.1.21.4):
  - Cleave within the recognition site;
  - e.g. BcgI and BplI, HindIII
- Type III enzymes (EC 3.1.21.5):
  - Cleave at sites a short distance from recognition site;
  - e.g. EcoP15, EcoP1I, EcoP15I
- Type IV enzymes:
  - Cleave at Target modified DNA - methylated, hydroxymethylated and glucosyl-hydroxymethylated DNA
  - e.g. EcoR124

Isoschizomer & Neoschizomers

- **Isoschizomer:**
  - **Restriction enzymes** specific to the same recognition sequence.
  - For example, SphI (CGTAC/G) and BbuI (CGTAC/G) are isoschizomers of each other.
  - The first enzyme discovered - Prototype; all subsequently identified - Isoschizomers.
  - Isoschizomers are isolated from different strains of bacteria and therefore may require different reaction conditions.
- **Neoschizomers**:
  - Recognizes the same sequence but cuts differently.
  - Specific type (subset) of isoschizomer.
  - For example, SmaI (CCC/GGG) and XmaI (C/CCGGG) are neoschizomers of each other

Isocaudomer

- Recognizes a slightly different sequence, but produces the same ends.
  - For example the enzymes Mbo I and BamH I are isocaudomers:
- Some Restriction Enzymes Used in Molecular Biology:

Enzyme	Restriction Site	Source
Aat I	G ACGT' C C' TGCA G	*Acetobacter aceti*
Acc III	T' CCGG A A GGCC' T	*Acinetobacter calcoaceticus*
*Bam*H I	G' GATC C C' CTAG' G	*Bacillus amyloliquetaciens*
*Eco*R I	G' AATT C C TTAA' G	*Escherichia coli* RY 13
*Eco*R V	GAT' ATC CTA' TAG	*Escherichia coli* J62 PLG74
Nco I	C' CATG G G GTAC' C	*Nocardia corallina*
Xba I	T' CTAG A A GATC' T	*Xanthomonas badrii*

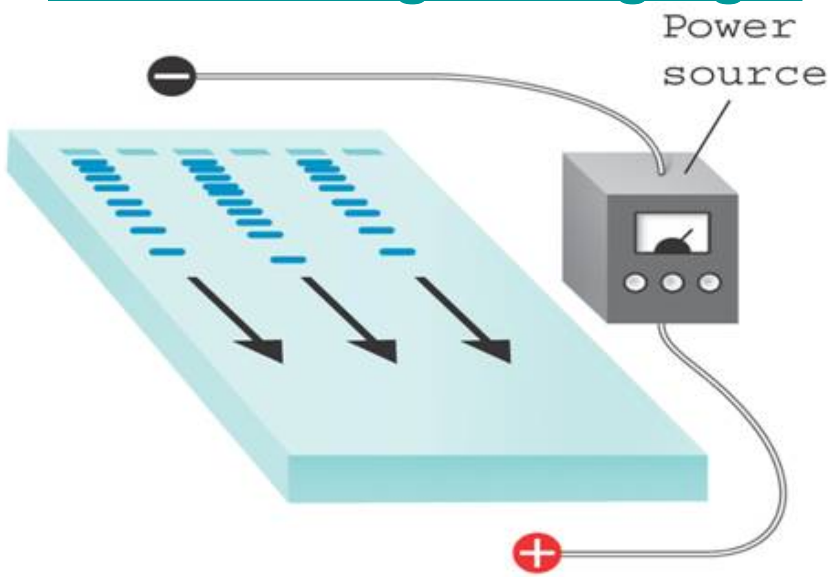
DNA Separation Techniques

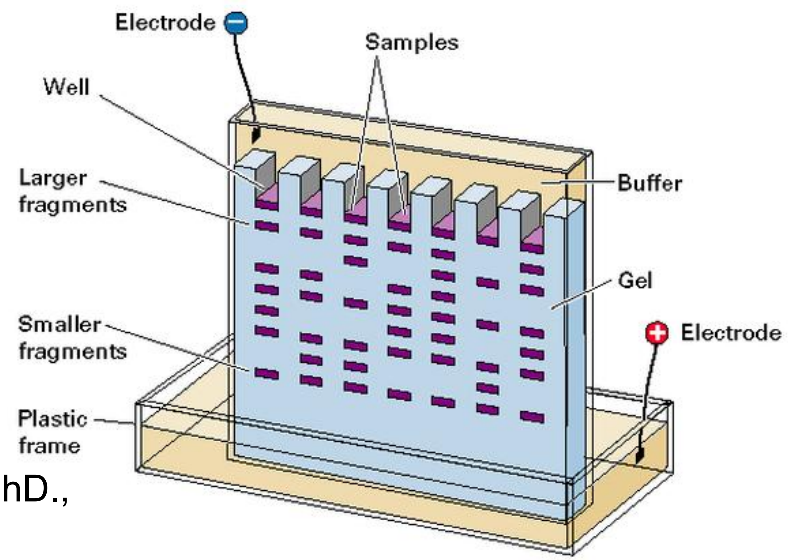
- **Separating DNA Molecules: Gel Electrophoresis and the Southern Blot**
  - Gel electrophoresis
    - Separates molecules based on electrical charge, size, and shape
    - Allows scientists to isolate DNA of interest
    - Negatively charged DNA drawn toward positive electrode
    - Agarose makes up gel; acts as molecular sieve
    - Smaller fragments migrate faster and farther than larger ones
    - Determine size by comparing distance migrated to standards

Gel Electrophoresis

- An electric current is applied to the gel to get the DNA moving
  - **Small molecules move faster**
  - **Big molecules move slower**
- The gel acts like a filter by separating strands of different sizes

▪ DNA moving through gel





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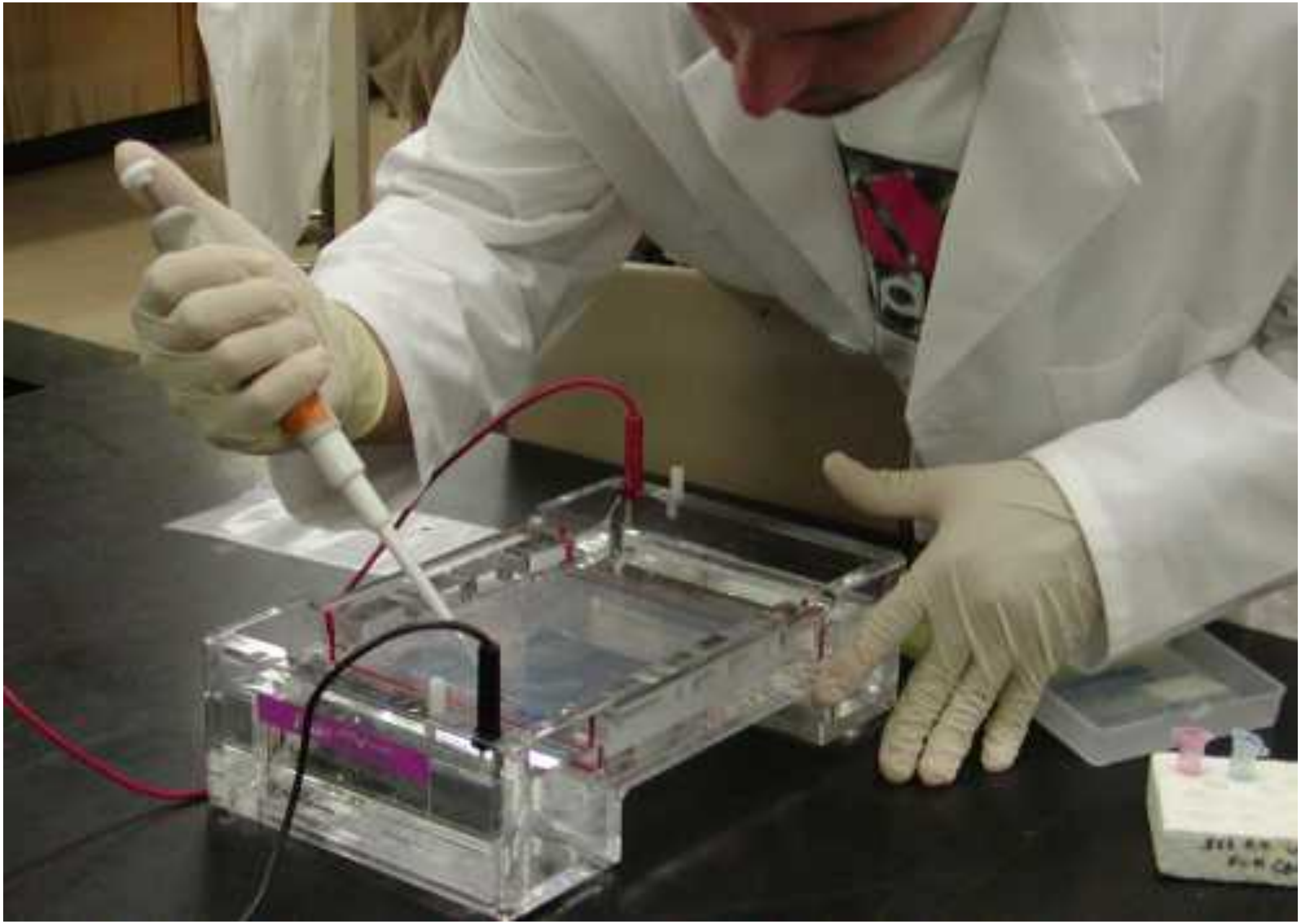
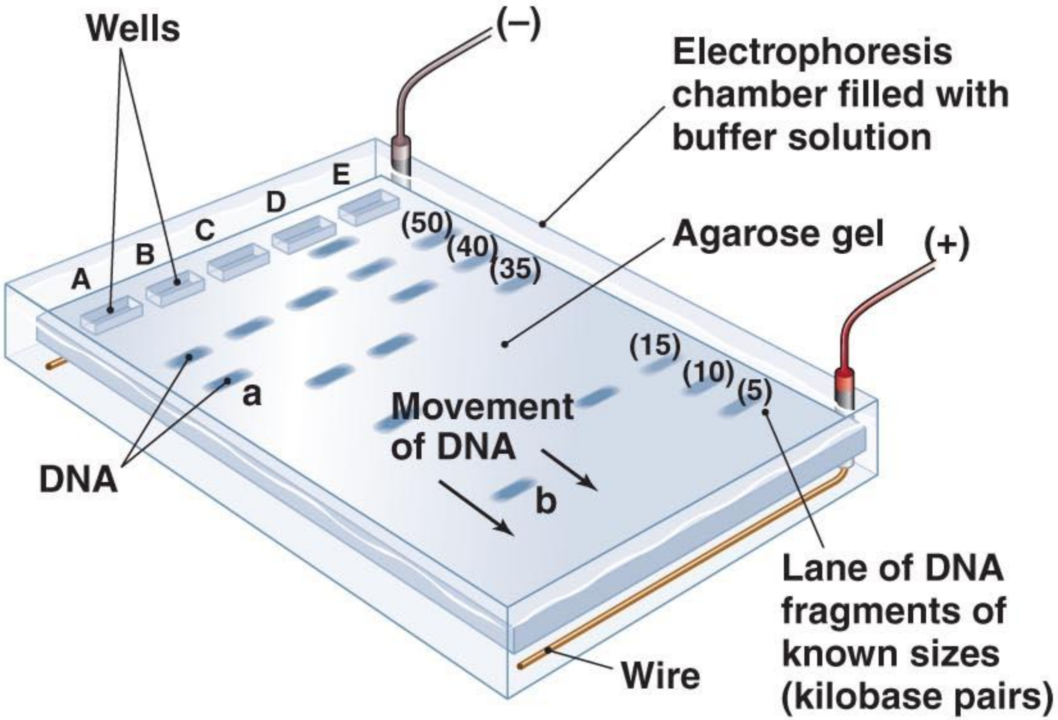
Gel Electrophoresis



Gel electrophoresis



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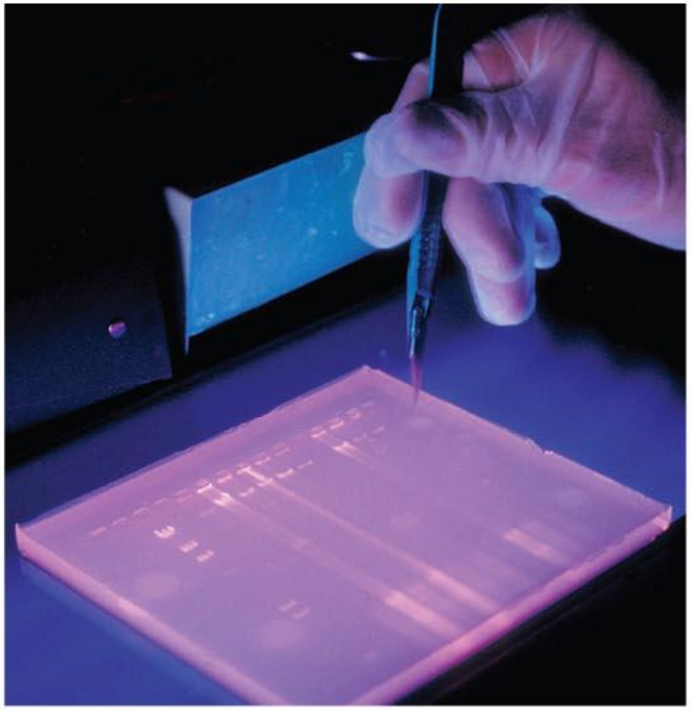

(b)

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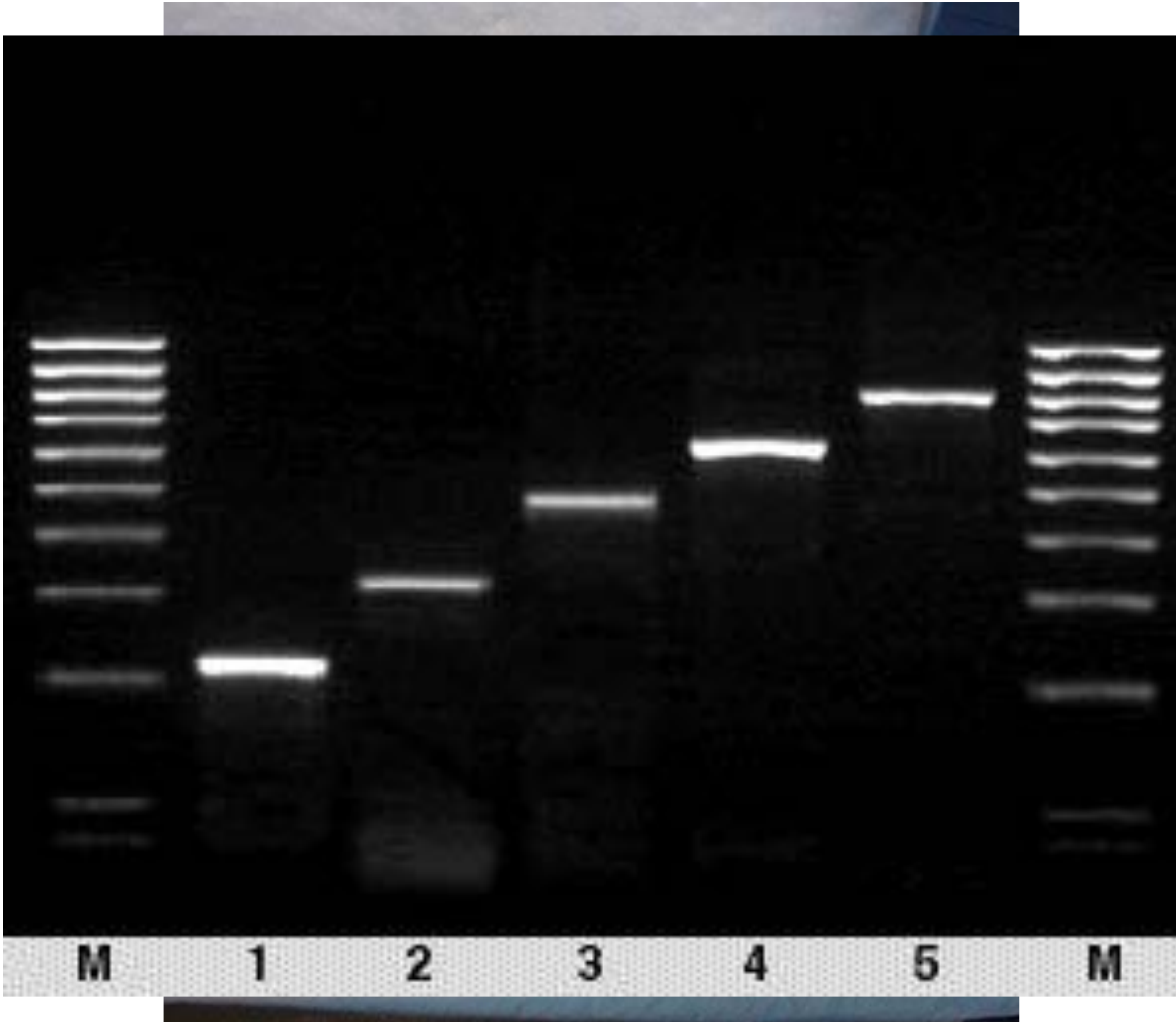
Figure 8.6

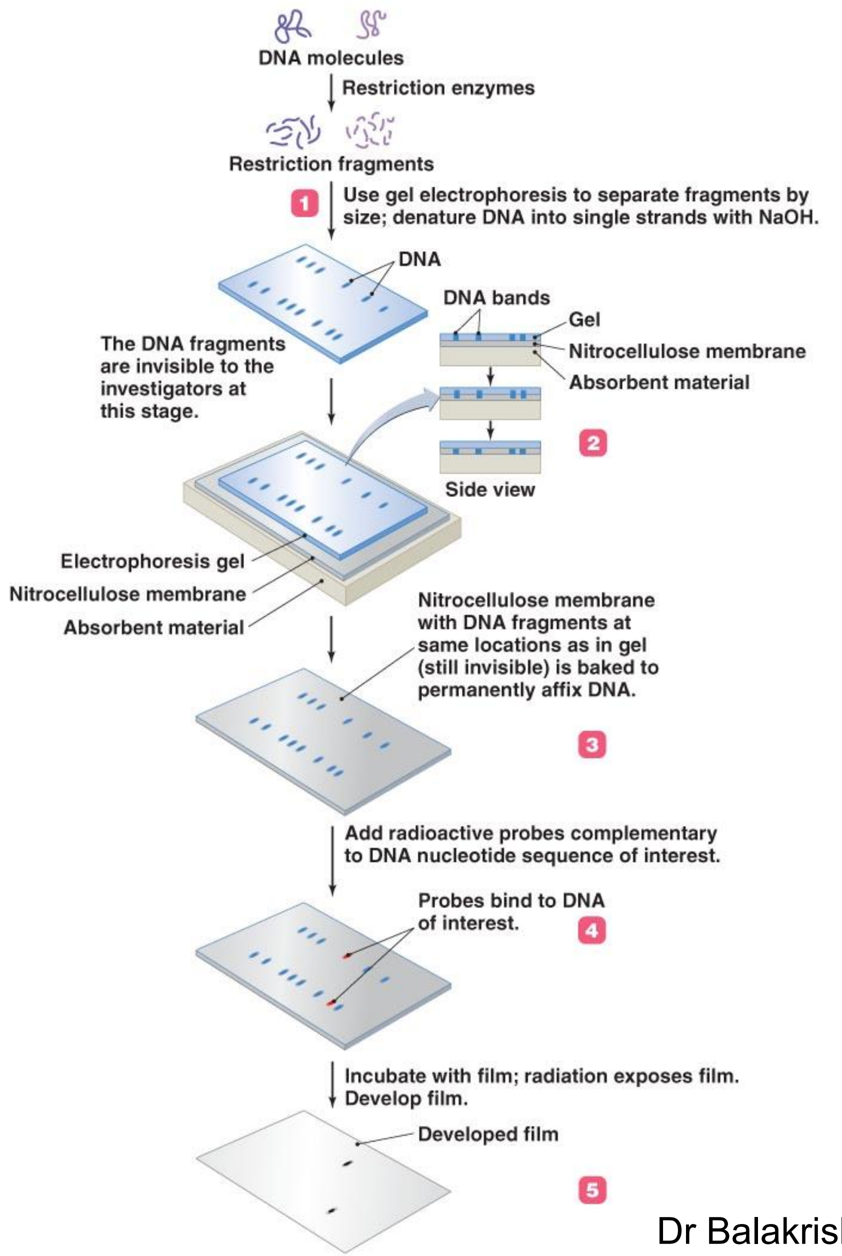
Gel Electrophoresis



DNA Separation Techniques

- **Separating DNA Molecules: the Southern Blot**
  - Southern blot
    - DNA transferred from gel to nitrocellulose membrane
    - Probes used to localize DNA sequence of interest
  - Uses of Southern blots
    - Genetic “fingerprinting”
    - Diagnosis of infectious disease

The Southern blot technique



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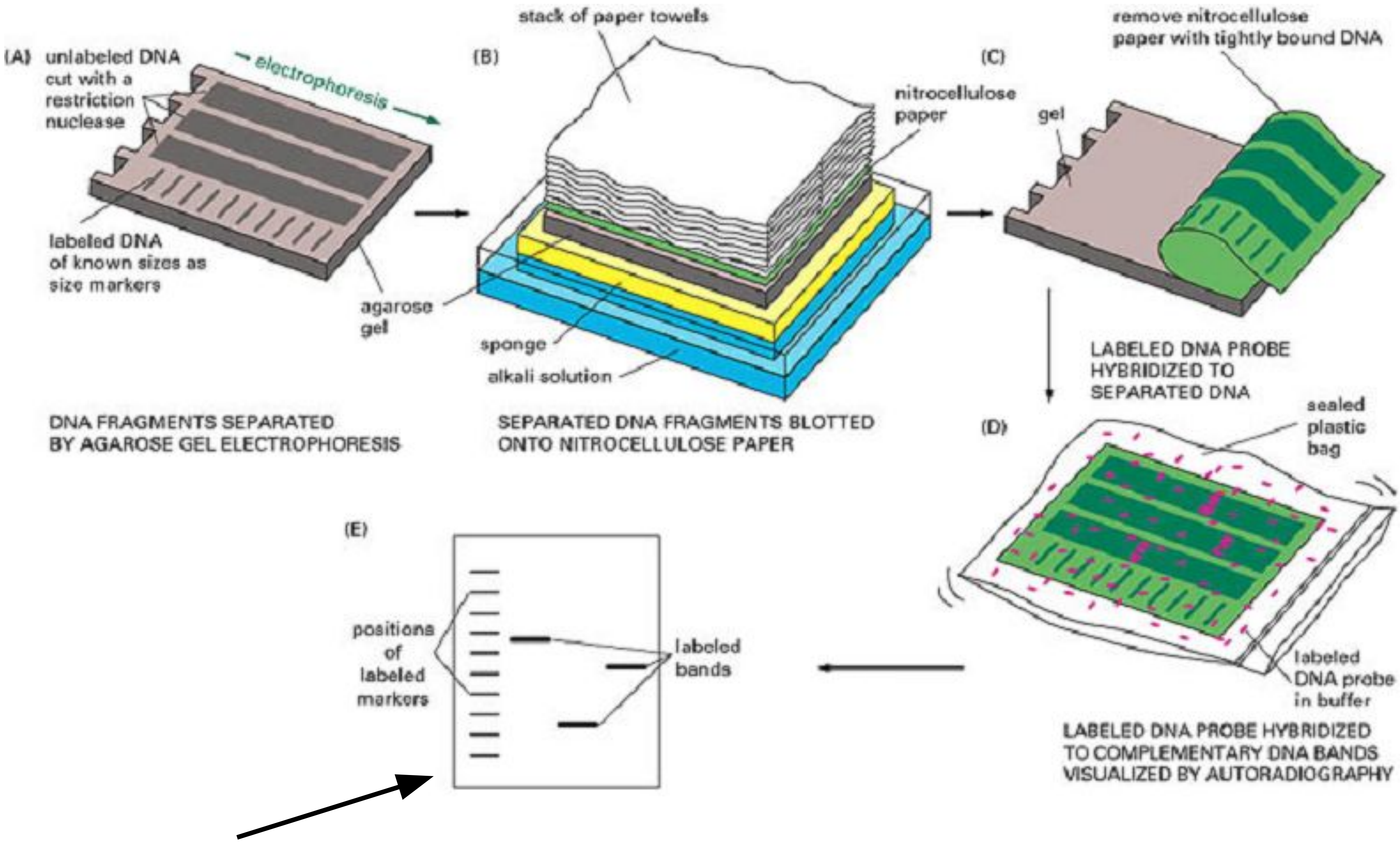
Southern Blotting



Visualized DNA

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Other Blotting techniques

- All techniques use electrophoresis to separate.
- Difference in techniques lies in the target
  - Western blotting:
    - Identify and locate proteins based on their ability to bind to specific antibodies
  - Northern blotting:
    - To detect specific RNA sequences using labeled probes.
  - Southwestern blotting:
    - Combines features of Southern and Western blotting techniques.
    - For rapid characterization of both DNA binding proteins and their specific sites on genomic DNA.

Comparison of Blotting Methods

	Southern	Northern	Western	Southwestern
What is separated	DNA cut with restriction enzymes	Denatured RNA	Protein denatured with SDS	Characterizes DNA binding proteins
Probe	Radioactive gene X DNA	Radioactive gene X DNA	Antibody against protein X, labeled with enzyme or radioactivity	Labelled DNA probes
What do you learn	Restriction map of gene X chromosome	How much gene X mRNA is present. How long is gene X mRNA	How much protein X is present. How large is protein X.	Identify expression of specific DNA binding proteins.

Nucleic Acid Detection

- DNA, RNA can be nonspecifically stained with ethidium bromide(EtBr) or SYBR (cyanine dye)

- DNA, RNA nucleic acids absorb light at 260 nm

  - Protein absorb light at 280 nm

  - 260/280 ratios to quantify the amount of nucleic acid (1.8 for pure DNA)

- Hybridization of a labeled nucleic acid to complementary sequences can identify specific nucleic acids.

  - **Probe –** A radioactive nucleic acid, DNA or RNA, used to identify a complementary fragment.

DNA probes:

- Probes are generally pieces of DNA or RNA labeled with a ^{32}P-containing nucleotide or fluorescently labeled nucleotides (more commonly now)
- Importantly, neither modification (^{32}P or fluorescent-label) affects the hybridization properties of the resulting labeled nucleic acid probes.
- The probe must recognize a complementary sequence to be effective.

Types of probes

- Most probes are single-stranded nucleic acid fragments complementary to the gene being sought
- Radioactive versus non-radioactive alternatives
  - Radioactive: Radioisotopes serve as the tag for identifying where the probe has bound to desired genomic or cDNA clones
    - Autoradiography required (X-ray film exposed to radioactivity)
  - Non-radioactive: Usually based on chemical reactions or color changes
    - Chemiluminescence
    - Colorimetric techniques
    - Fluorescence (Fluorescence in situ hybridization = FISH)

Types of probes

- Oligonucleotide probes
  - Short synthetic ssDNA
- RNA probes
  - Generated via in vitro transcription with RNA polymerase from SP6 or T7 promoter
- Antibodies
  - Used for “expression” libraries (lambda gt11)
  - Fusion proteins (beta galactosidase + cDNA product)

Sources of probes

- Sources of probes
  - Heterologous probes
    - From another species (provided genes are highly conserved)
    - “Phone-a-clone”
  - cDNA probes
    - To recover genomic sequences when introns and promoter elements are needed
  - Probe based on protein sequence
    - If the amino acid composition of a protein is known, then degenerate oligonucleotide probes can be generated
    - 18-21 bases is sufficient for specific probe (6-7 aa)

Nucleic Acid Detection

- **Autoradiography –** A method of capturing an image of radioactive materials on film.

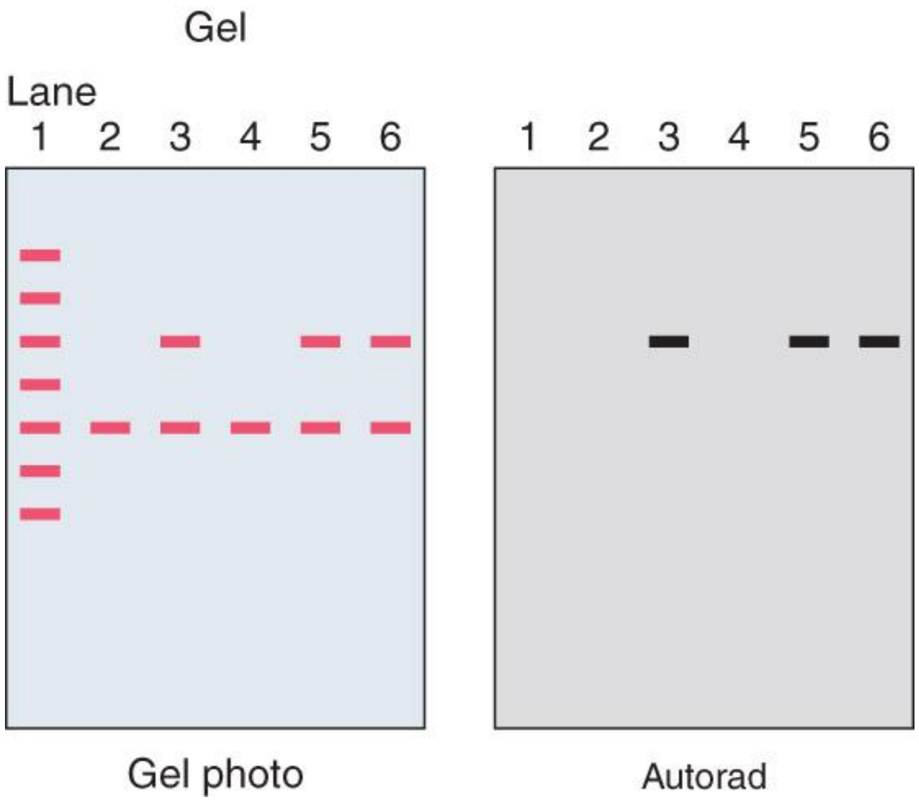

FIGURE: An autoradiogram of a gel prepared from different sample

Nucleic Acid Detection - FISH

- ***in situ* hybridization –** Hybridization of a probe to intact tissue to locate its complementary strand by autoradiography.

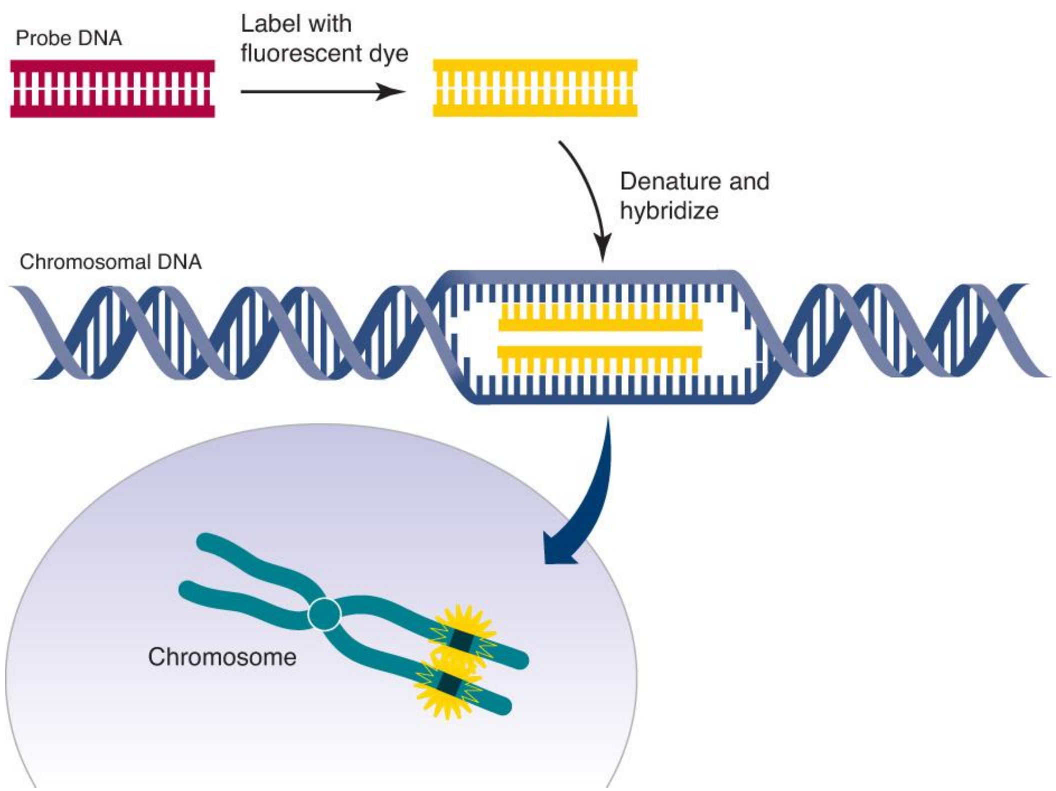

FIGURE: The fluorescent in situ hybridization (FISH) technique

Adapted from an illustration by Darryl Leja, National Human Genome Research Institute (www.genome.gov).

Fluorescence in situ Hybridization (FISH)

- FISH - a process which vividly paints chromosomes or portions of chromosomes with fluorescent molecules
- Identifies chromosomal abnormalities
- Aids in
  - Gene mapping,
  - Toxicological studies,
  - Analysis of chromosome structural aberrations,
  - Ploidy determination

In situ Hybridization: Locating genes in chromosomes

FISH:
Fluorescence
***in situ* hybridization**

Figure 5.17 Using a fluorescent probe to find a gene in a chromosome by in situ hybridization.

From Dr. Yu

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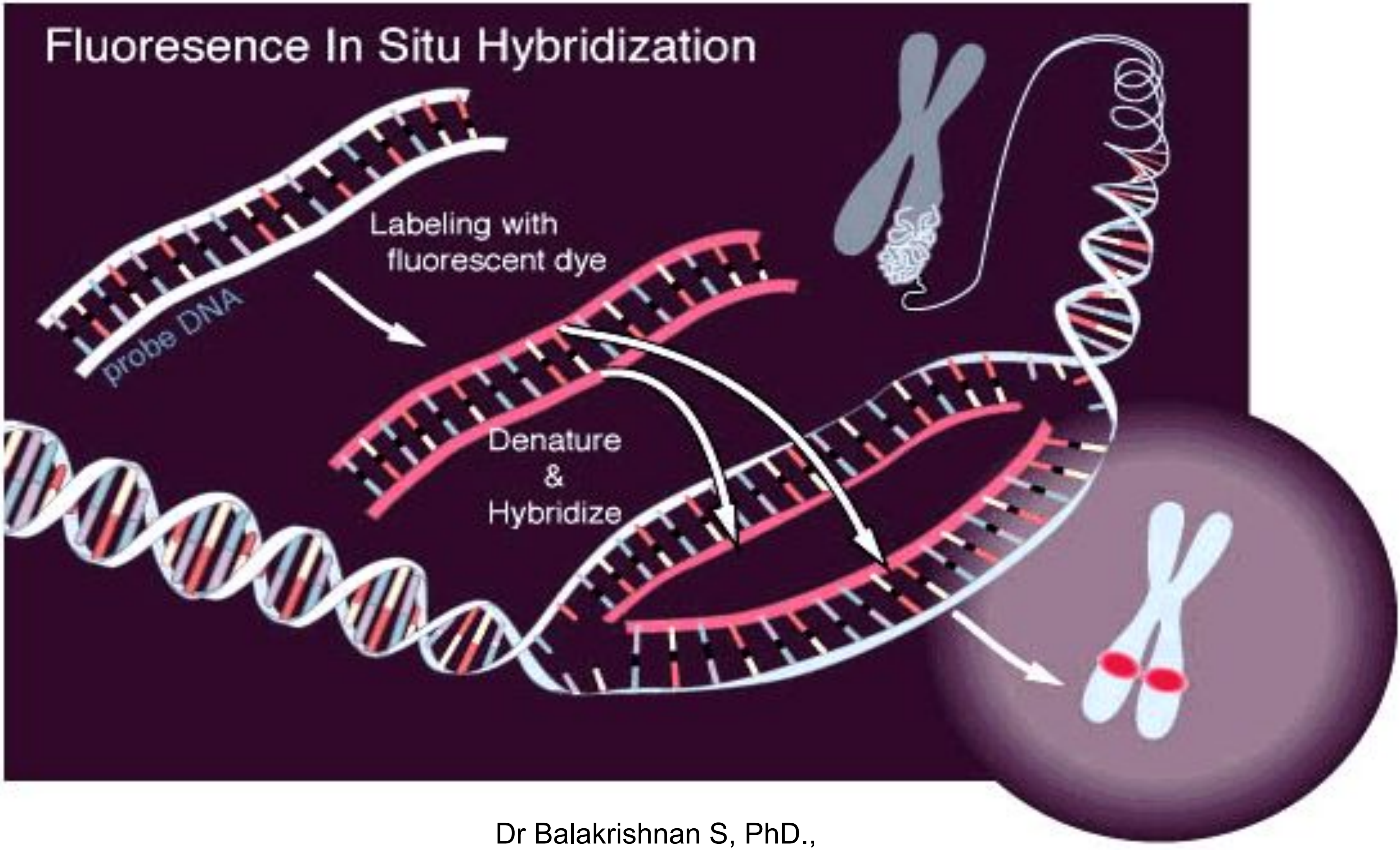
FISH Procedure

- Denature the chromosomes
- Denature the probe
- Hybridization
- Fluorescence staining
- Examine slides or store in the dark

FISH Procedure



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DNA Sequencing

- Process of determining the precise order of nucleotides within a DNA molecule
- First DNA sequences were obtained in the early 1970s
- Knowledge of DNA sequences has become indispensable for basic biological research
- Applications are;
  - Molecular biology: to study genomes and the proteins they encode
  - Evolutionary biology: to study how different organisms are related and evolved
  - Metagenomics: to know which organisms are present in a particular environment
  - Medicine: to determine if there is risk of genetic diseases
  - Forensics: forensic identification and paternity testing

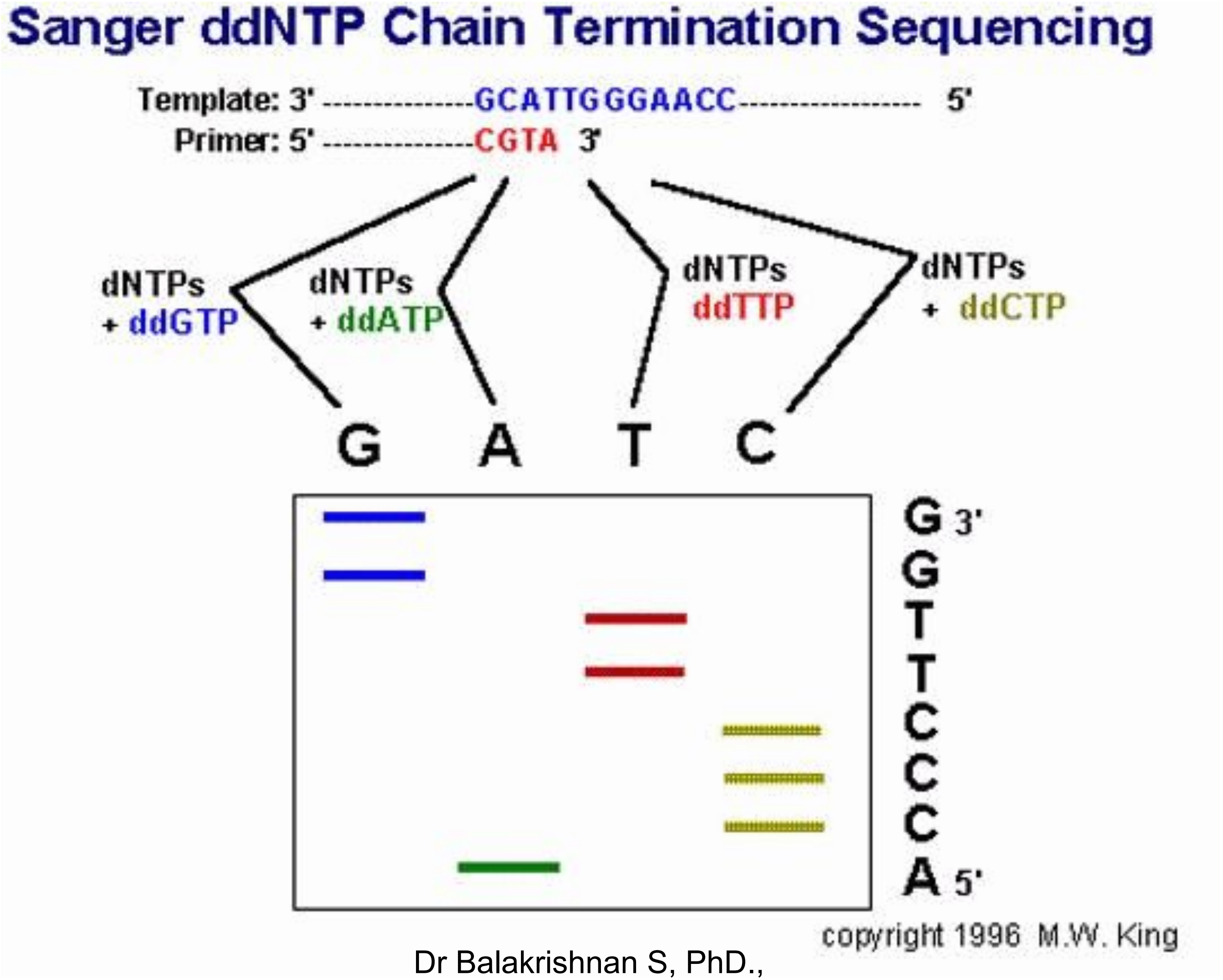
DNA Sequencing

- **Basic Methods:**
  1. Enzymatic method – Sanger's ("DNA sequencing with chain-terminating inhibitors")
  2. Chemical method – Maxam & Gilbert ("DNA sequencing by chemical degradation")
- **Advanced methods and de novo sequencing:**
  - Shotgun sequencing
  - Bridge PCR
- **High-throughput methods:**
  - Massively parallel signature sequencing (MPSS)
  - Polony sequencing
  - Illumina (Solexa) sequencing
  - Combinatorial probe anchor synthesis (cPAS)
  - SOLiD sequencing
  - Ion Torrent semiconductor sequencing
  - DNA nanoball sequencing
  - Heliscope single molecule sequencing
  - Single molecule real time (SMRT) sequencing
  - Nanopore DNA sequencing

DNA Sequencing – enzymatic method

- "DNA sequencing with chain-terminating inhibitors"
- Developed by Fredrick Sanger and colleagues in 1977.
- Most widely used ~40 years.
- Based on the selective incorporation of chain terminating ddNTPs by DNA pol during in vitro DNA rep.
- Requires:
  - ssDNA template
  - DNA primer
  - DNA pol.
  - dNTPs
  - ddNTPs (lack 3'-OH group)
- Chain termination sequencing uses **dideoxynucleotides** (ddNTPs) to terminate DNA synthesis at particular nucleotides.
- The ddNTP concentration should be ~100-fold lower than that of corresponding dNTP to allow enough fragments to be produced.
- **Primer -** A single stranded nucleic acid molecule with a 3′ –OH used to initiate DNA polymerase replication of a paired template strand.

DNA Sequencing



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DNA Sequencing

DNA sequence

A G C G T A T C G G
1 2 3 4 5 6 7 8 9 10

A* terminated:

A*
1

A*
1 2 3 4 5 6

G* terminated:

G*
1 2

G*
1 2 3 4

G*
1 2 3 4 5 6 7 8 9

G*
1 2 3 4 5 6 7 8 9 10

C* terminated:

C*
1 2 3

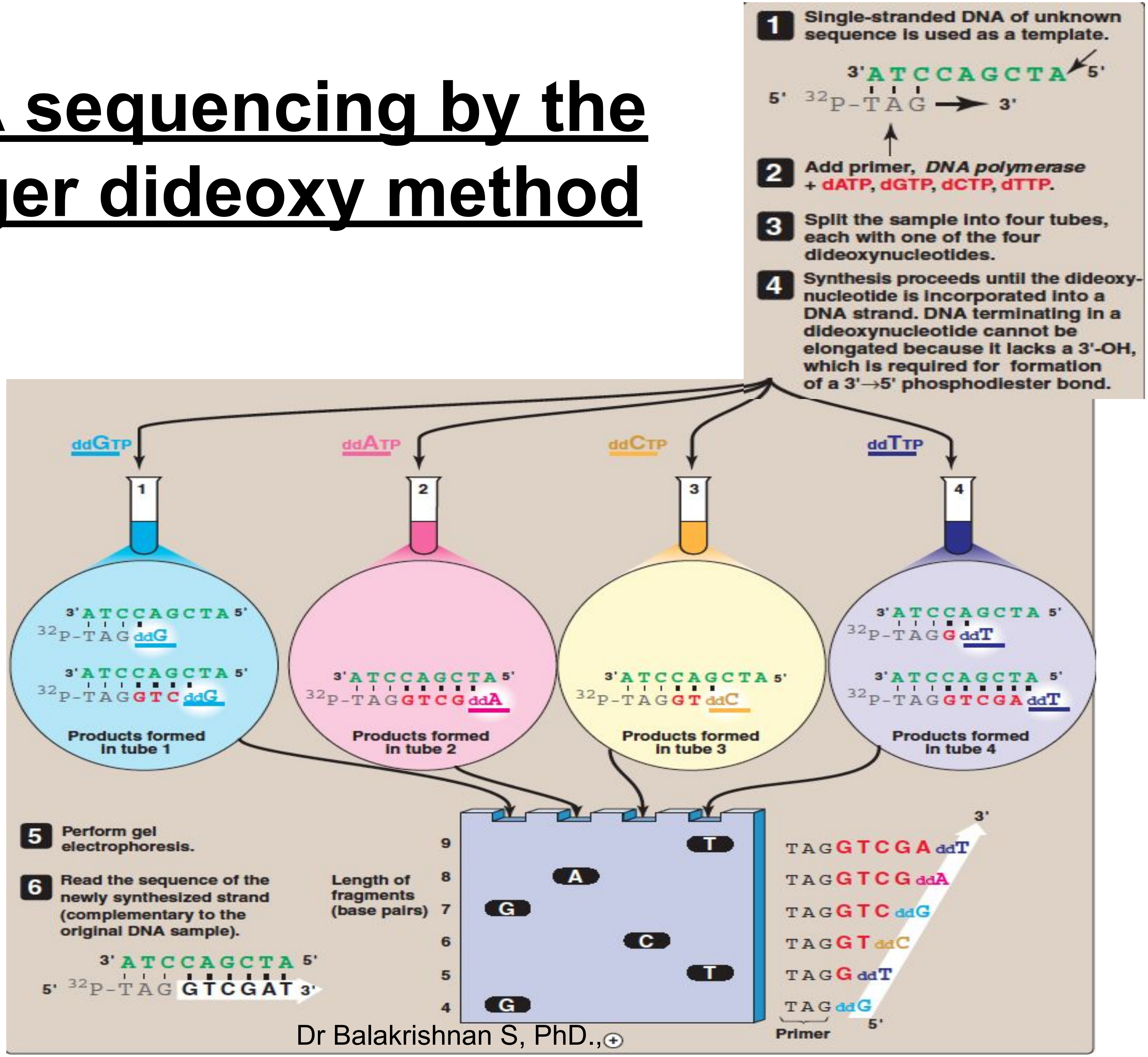
C*
1 2 3 4 5 6 7 8

T* terminated:

T*
1 2 3 4 5

T*
1 2 3 4 5 6 7

DNA sequencing by the Sanger dideoxy method

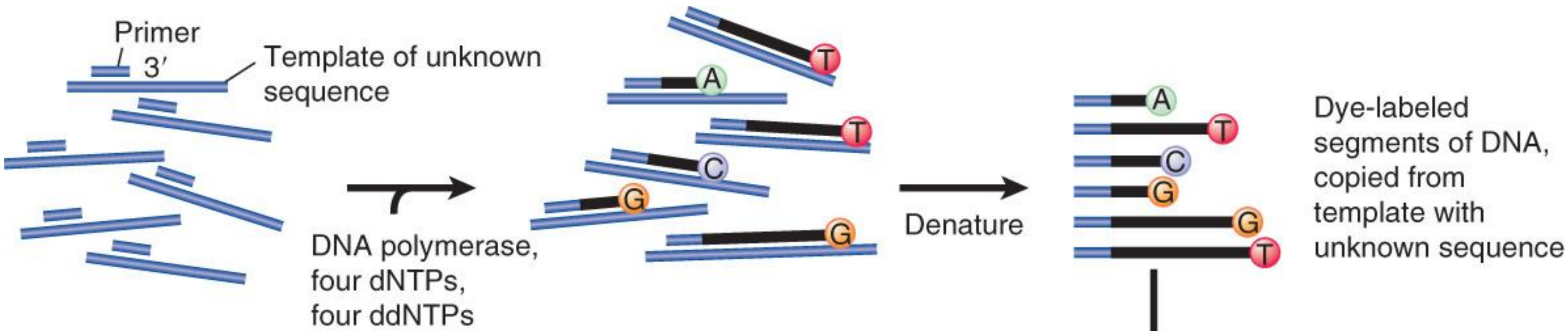


DNA Sequencing

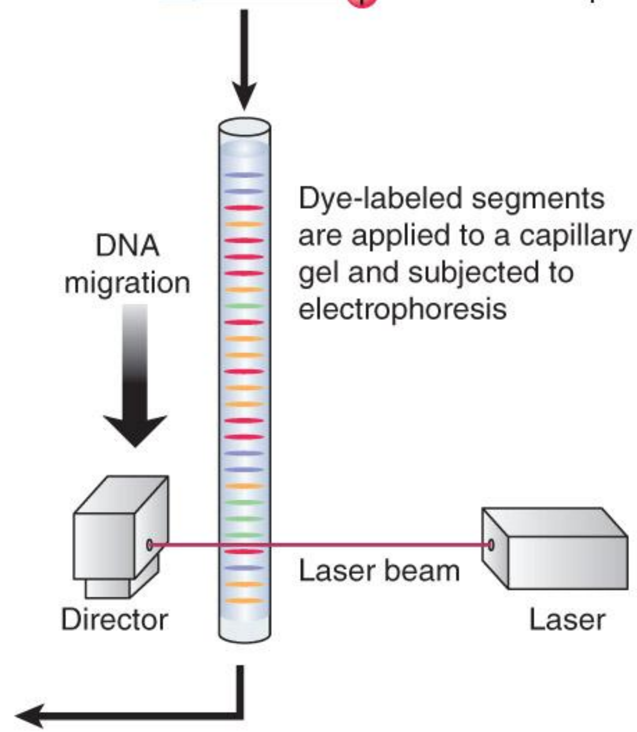
- Chain termination sequencing uses ddNTPs to terminate DNA synthesis at particular nucleotide.

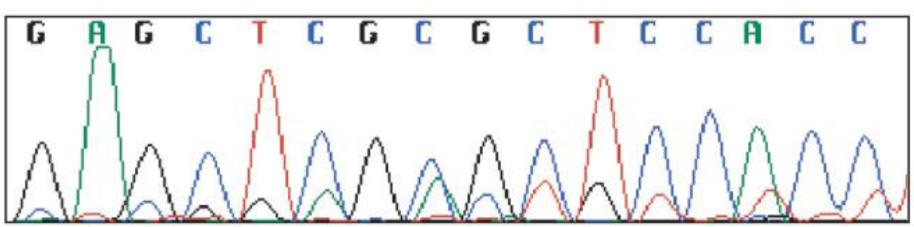
- Fluorescently tagged ddNTPs and capillary gel electrophoresis allow automated, high-throughput DNA sequencing (Leory Hood and coworkers).

FIGURE: DideoxyNTP sequencing using fluorescent tags



1. Different fluorescents label for each ddNTP
2. Hit with a laser and pass by an optical sensor
3. Glass capillary tube dissipate heat rapidly





Computer-generated result after bands migrate past detector

Inset photo courtesy of Jan Kieleczawa

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DNA Sequencing

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Automated DNA sequencing

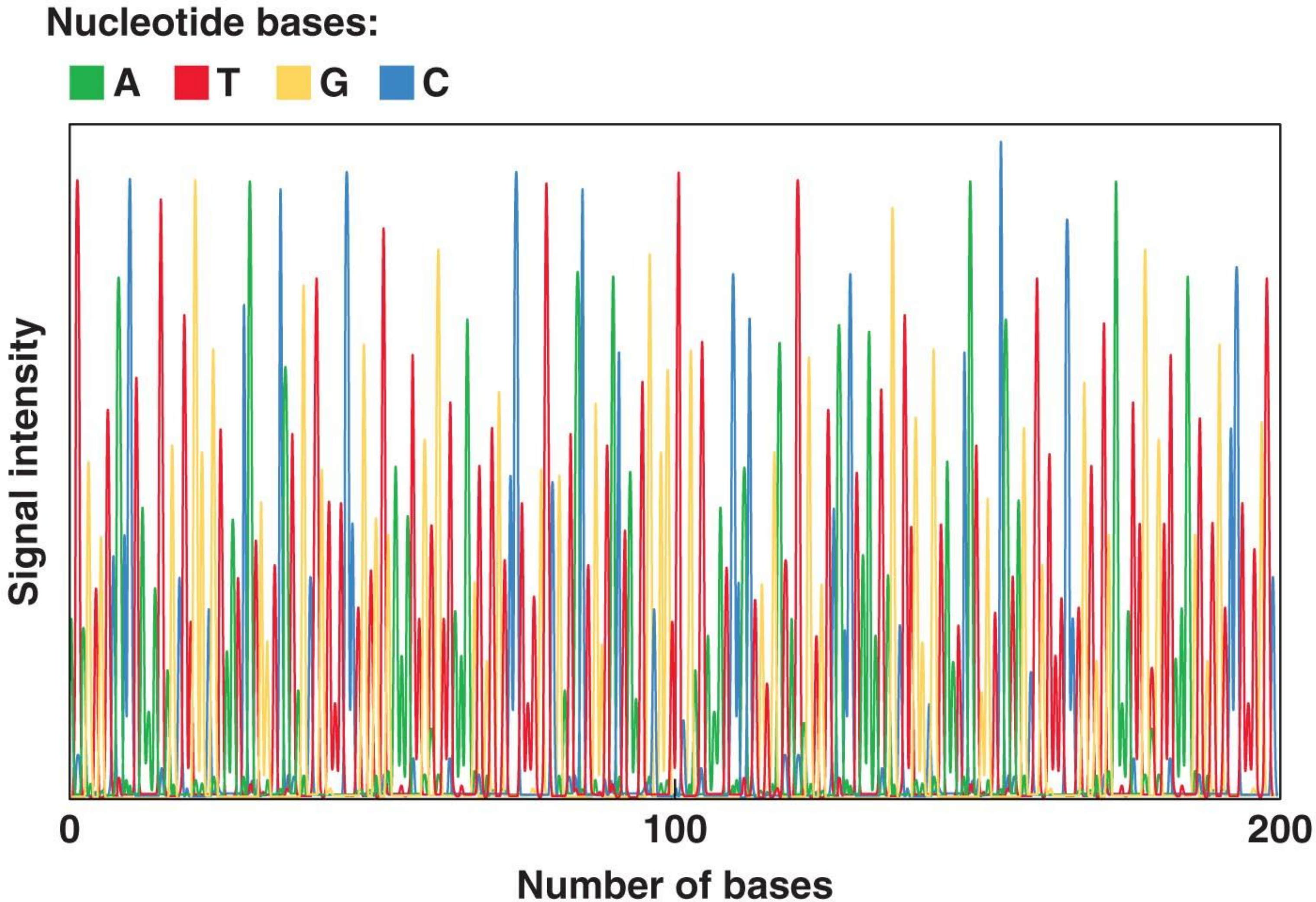

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Figure 8.11

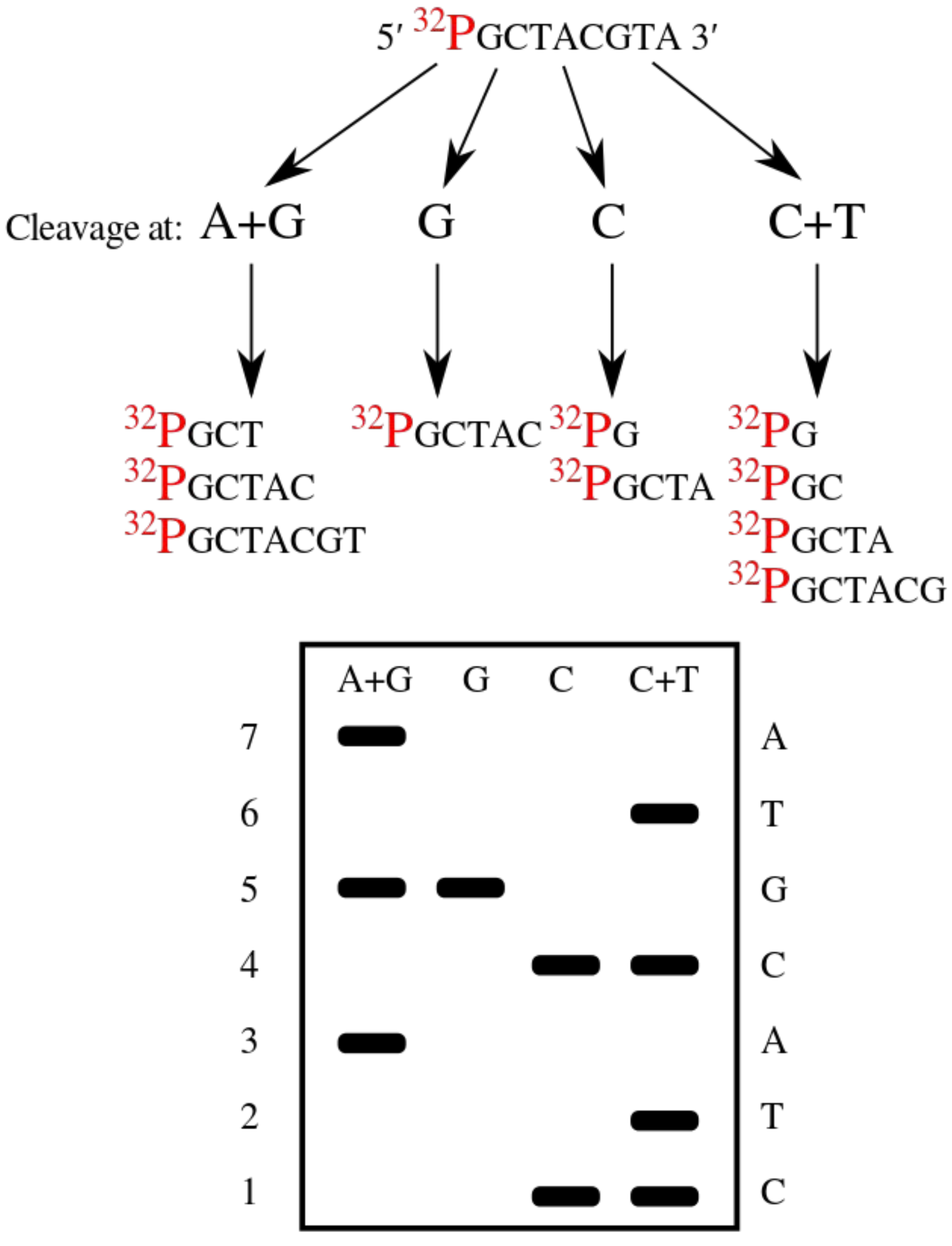
DNA Sequencing – Chemical method

- Proposed by Allan Maxam and Walter Gilbert in 1976 - 1977.
- First widely adopted method
- Based on nucleobase-specific partial chemical modification of DNA and subsequent cleavage of DNA backbone at site adjacent to modified nucleotides.
- Use of radioactive labeling and its technical complexity discouraged extensive use after refinements in the Sanger methods had been made.
- Steps:
  - Requires radioactive labeling at one 5' end of the DNA and purification of the DNA fragment to be sequenced.
  - Chemical treatment then generates breaks (using hot piperidine) at a small proportion of one or two of the four nucleotide bases in each of four reactions (G, A+G, C, C+T).

DNA Sequencing – Chemical method

- The chemical treatments are:
  - Depurination of purines (A+G) using formic acid,
  - Methylation of guanine (G) by dimethy sulfate,
  - Hydrolysis of pyrimidines (C+T) using hydrazine, and
  - Addition of salt to hydrazine reaction to inhibit reaction of (T) for (C) only reaction
- The concentration of the modifying chemicals is controlled to introduce on average one modification per DNA molecule.
- Thus a series of labeled fragments is generated, from the radiolabeled end to the first "cut" site in each molecule.
- The fragments in the four reactions are electrophoresed side by side in denaturing acrylamide gels for size separation.
- To visualize the fragments, the gel is exposed to X-ray film for autoradiography, yielding a series of dark bands each corresponding to a radiolabeled DNA fragment, from which the sequence may be inferred.

DNA Sequencing – Chemical method



Sequencing Gel

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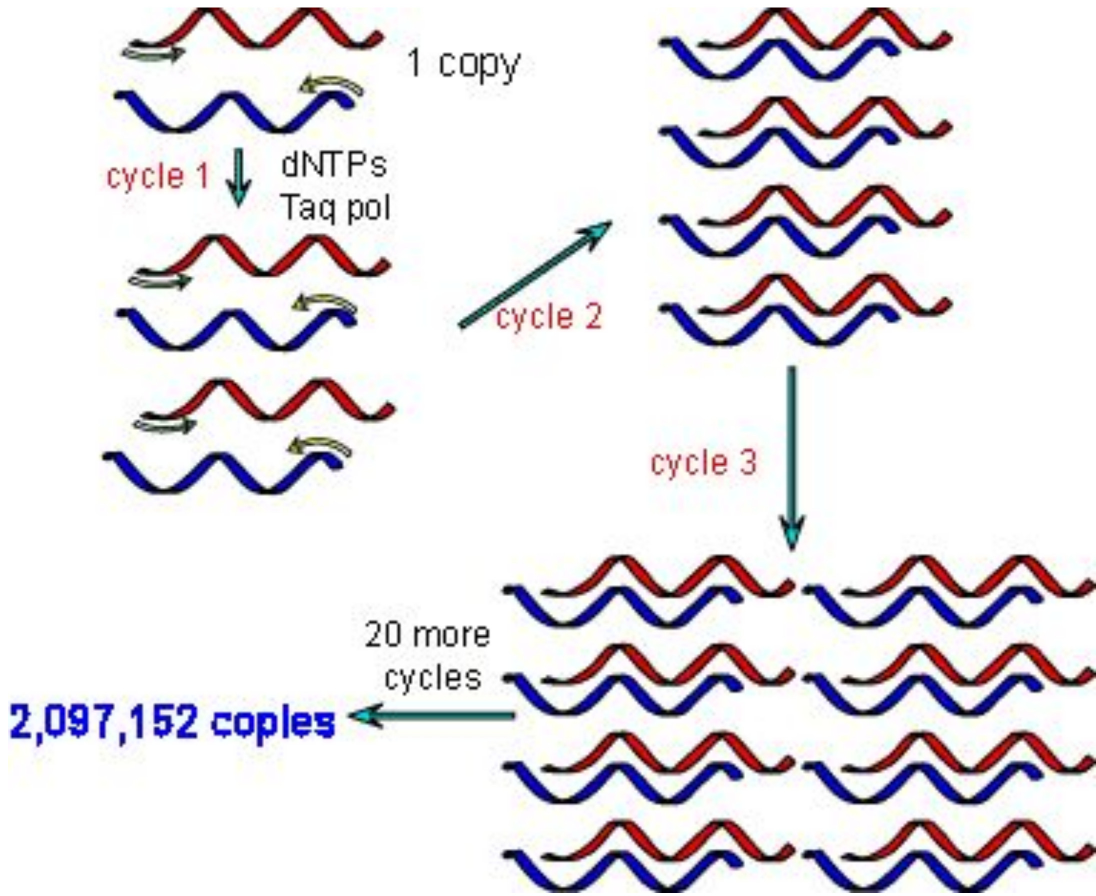
How much DNA is needed?

- To perform DNA fingerprinting, you need a lot of DNA
- Many copies of the DNA can be made using **polymerase chain reaction**

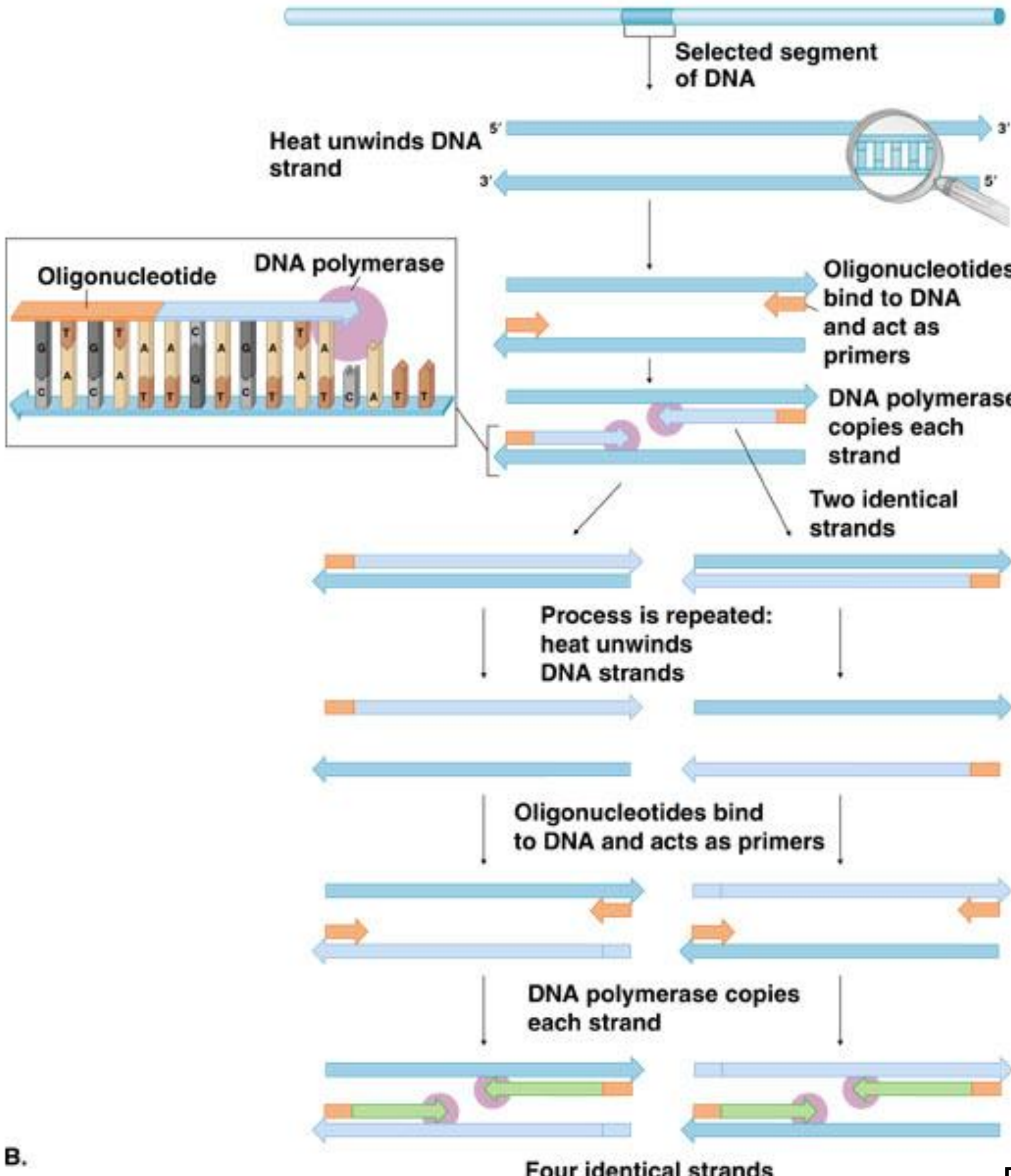


PCR – polymerase chain reaction

Now we can easily copy or **amplify** DNA in the lab.

1. Produce two oligonucleotides that match up with the DNA you want to copy.
2. Heat DNA to break the hydrogen bonds and open the DNA (94°C)
3. Add oligonucleotides (primers)
4. Cool, allowing primers to bind to DNA (60°C)
5. Add nucleotides and DNA polymerase. (72°C)
6. Repeat

PCR – Mechanism



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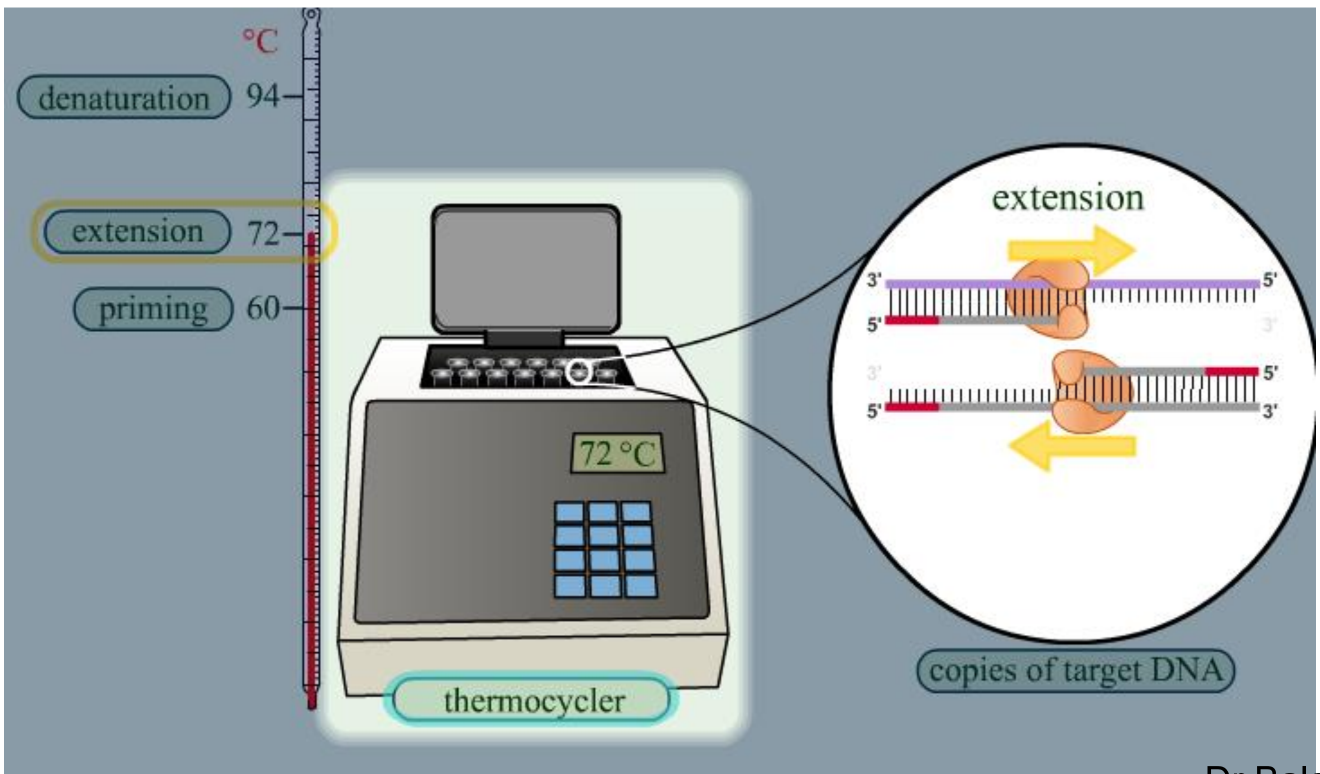
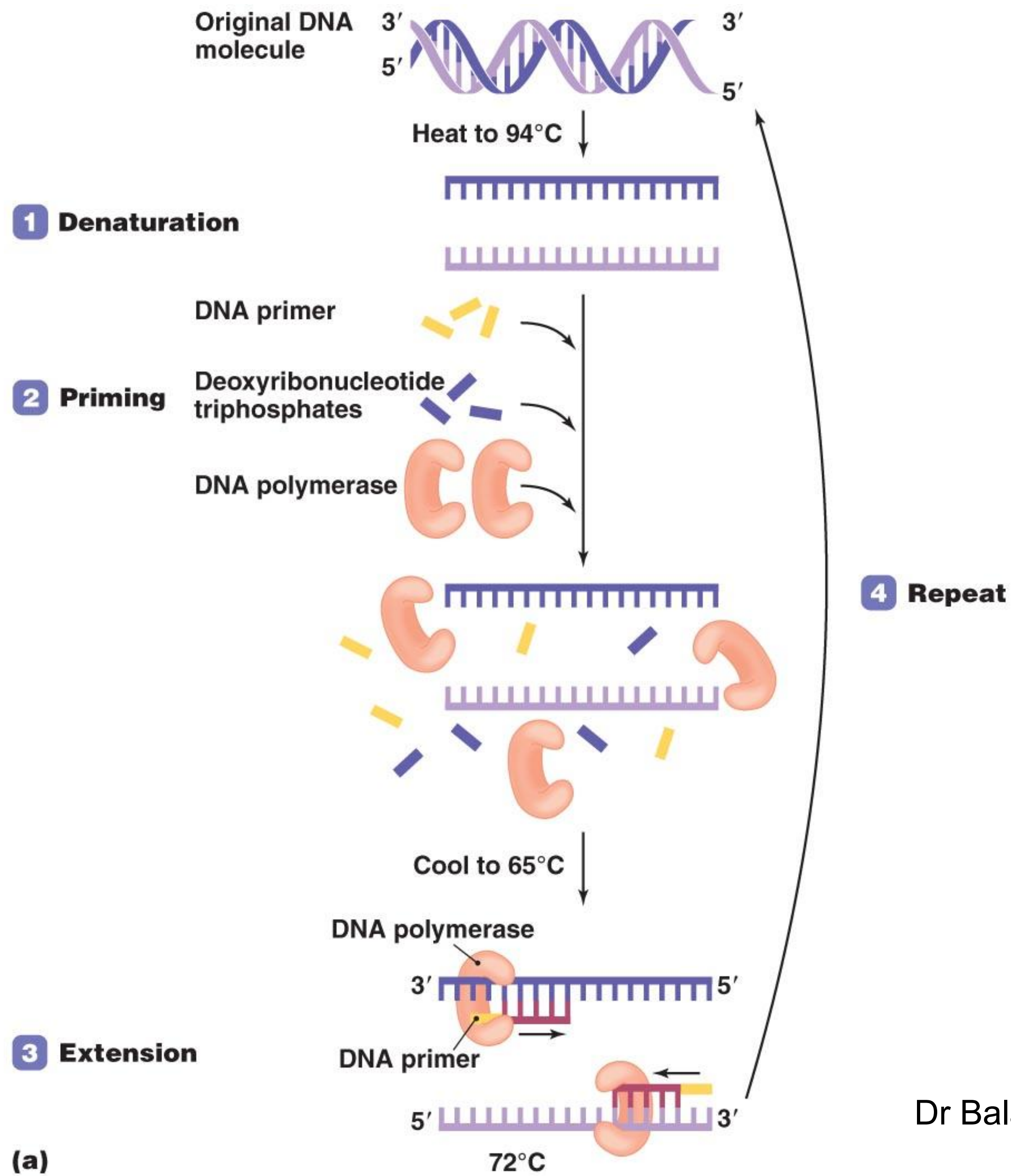
PCR: The Process

- Repetitive process consisting of three steps
  - Denaturation
  - Priming
  - Extension
- Can be automated using a thermocycler



Polymerase chain reaction (PCR)



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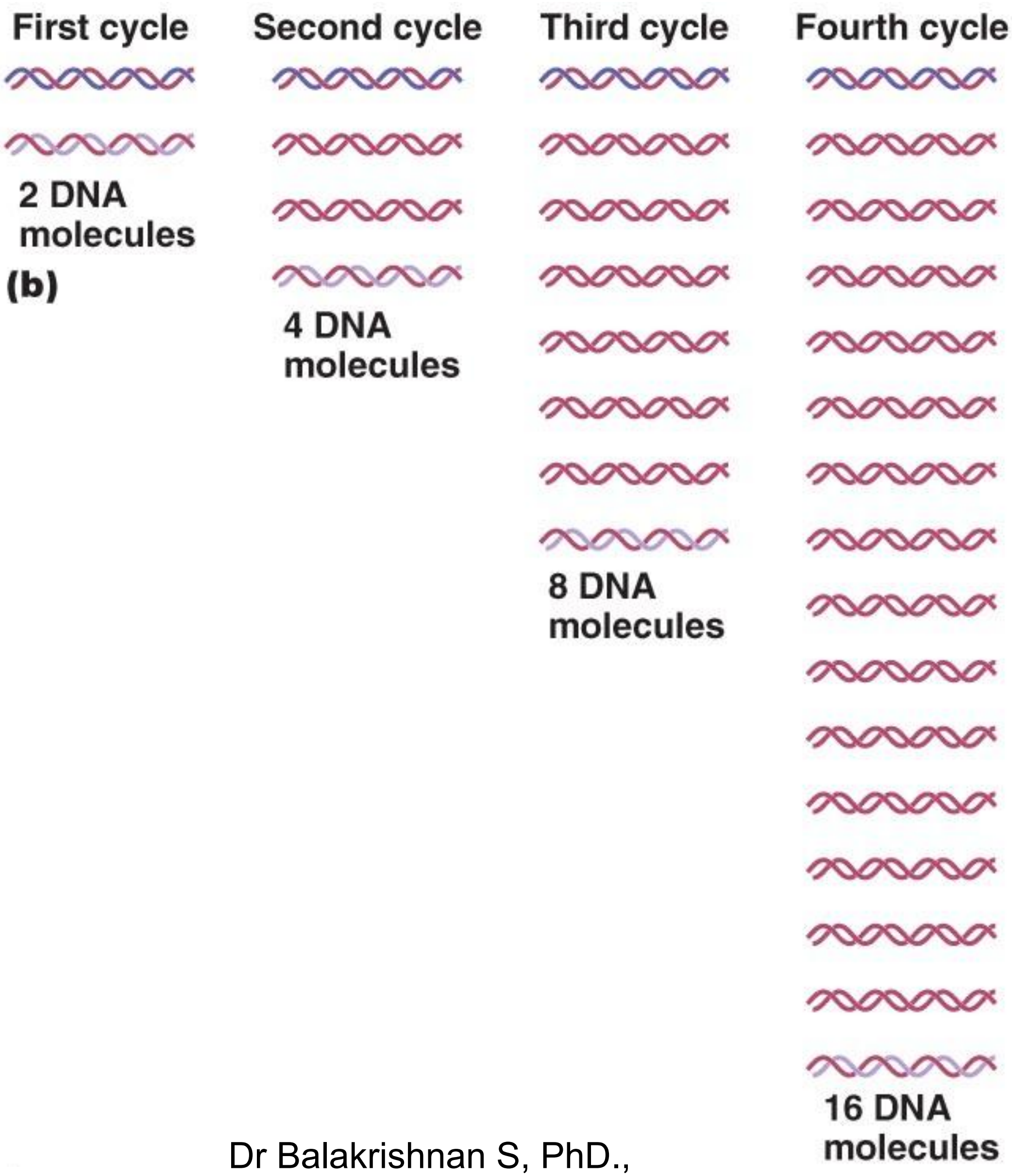
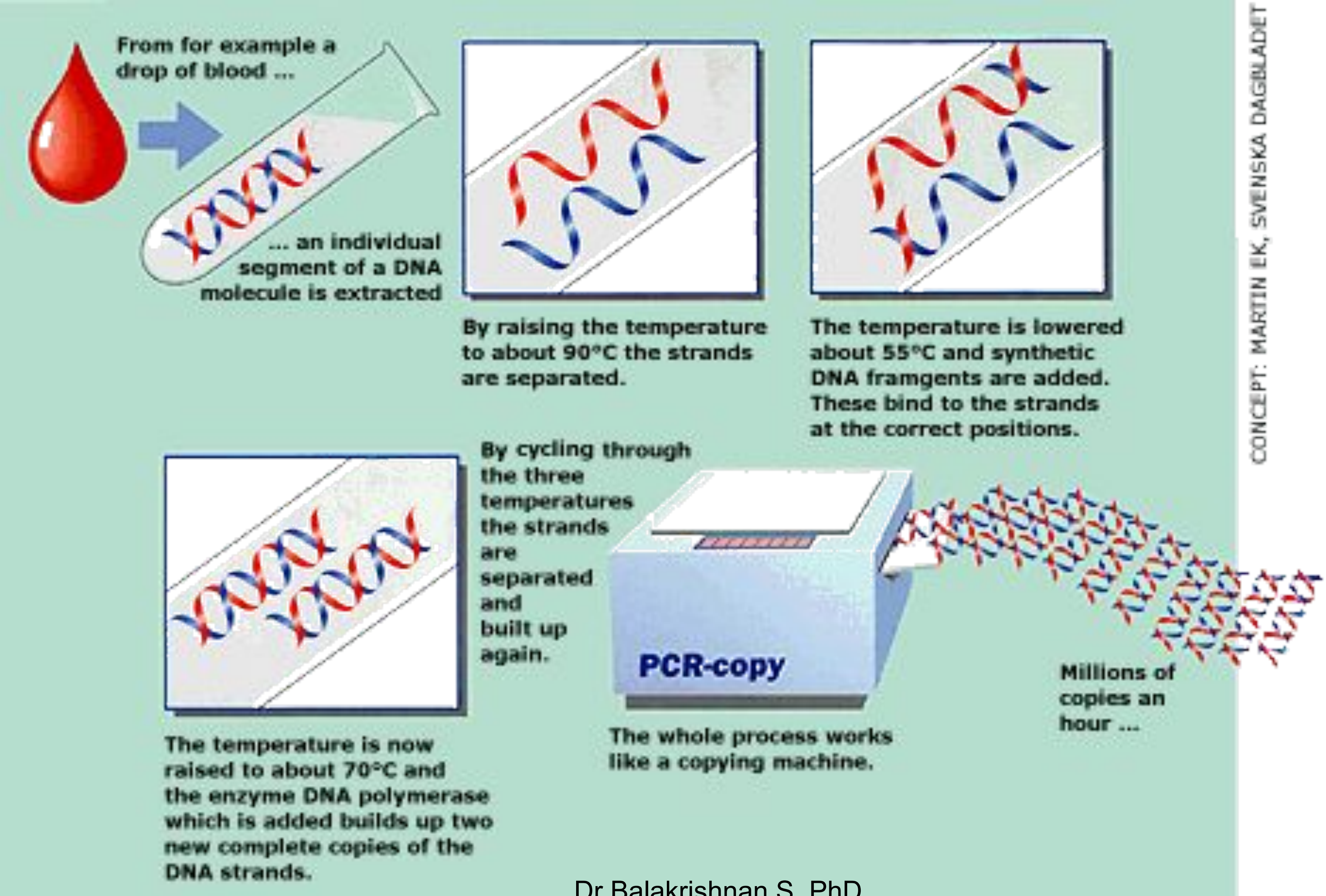
Polymerase chain reaction (PCR)



PCR



From for example a drop of blood ...
... an individual segment of a DNA molecule is extracted
By raising the temperature to about 90°C the strands are separated.
The temperature is lowered about 55°C and synthetic DNA framgents are added. These bind to the strands at the correct positions.
By cycling through the three temperatures the strands are separated and built up again.
PCR-copy
Millions of copies an hour ...
The temperature is now raised to about 70°C and the enzyme DNA polymerase which is added builds up two new complete copies of the DNA strands.
The whole process works like a copying machine.
CONCEPT: MARTIN EK, SVENSKA DAGBLADET

Application of PCR

- PCR can be applied to :
  - Amplifying DNA in a drop of blood
  - Find DNA from the HIV virus
  - Find early signs of cancer
  - Find genetic defects in human embryos
  - Examine the DNA of ancient organisms

PCR and RT-PCR

- **RT-PCR** uses reverse transcriptase to convert RNA to cDNA for use in a PCR reaction.

- **Real-time**, or **quantitative**, **PCR** detects the products of PCR amplification during their synthesis, and is more sensitive and quantitative than conventional PCR.

Cutting DNA

- Everyones DNA sequences are unique
  - Recognition sequences are in different places on different people
  - When a restriction enzyme cuts the DNA of 2 different people, it will cut it into pieces that are different sizes

Suspect #1

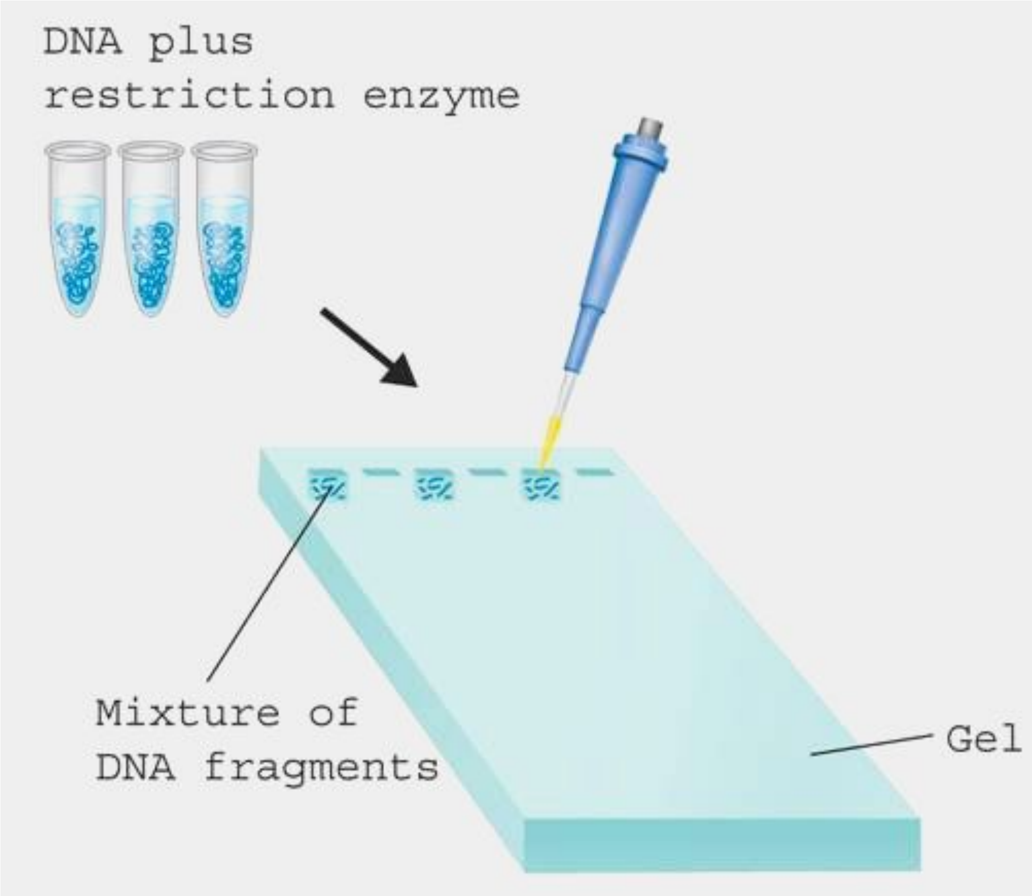
Suspect #2

Analyzing DNA

- The pieces are then analyzed
  - Each piece has its own unique size and shape
- These properties are used to perform **DNA fingerprinting**

DNA fingerprinting

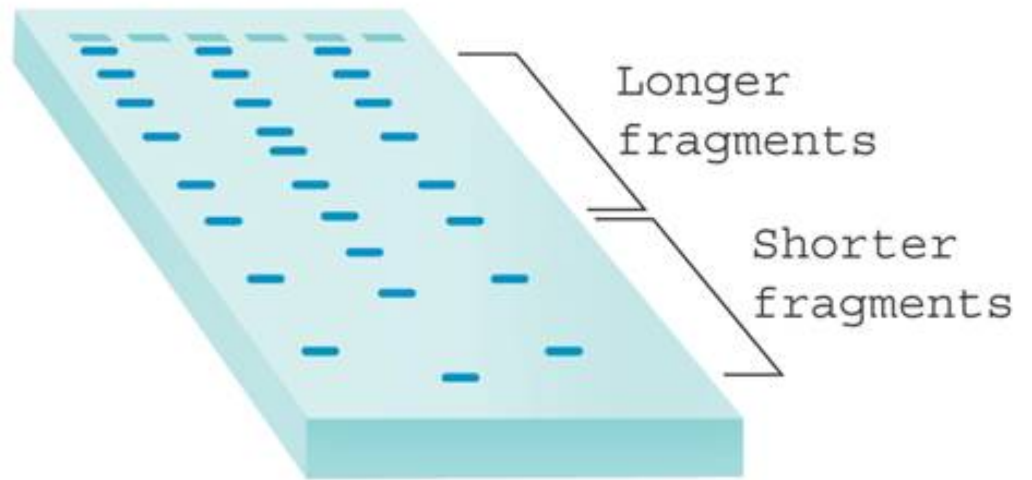
- To get the genetic "fingerprint" of an organism, its DNA is first cut using restriction enzymes
- The mixture of DNA and enzymes are then placed into a gel to be separated by gel electrophoresis



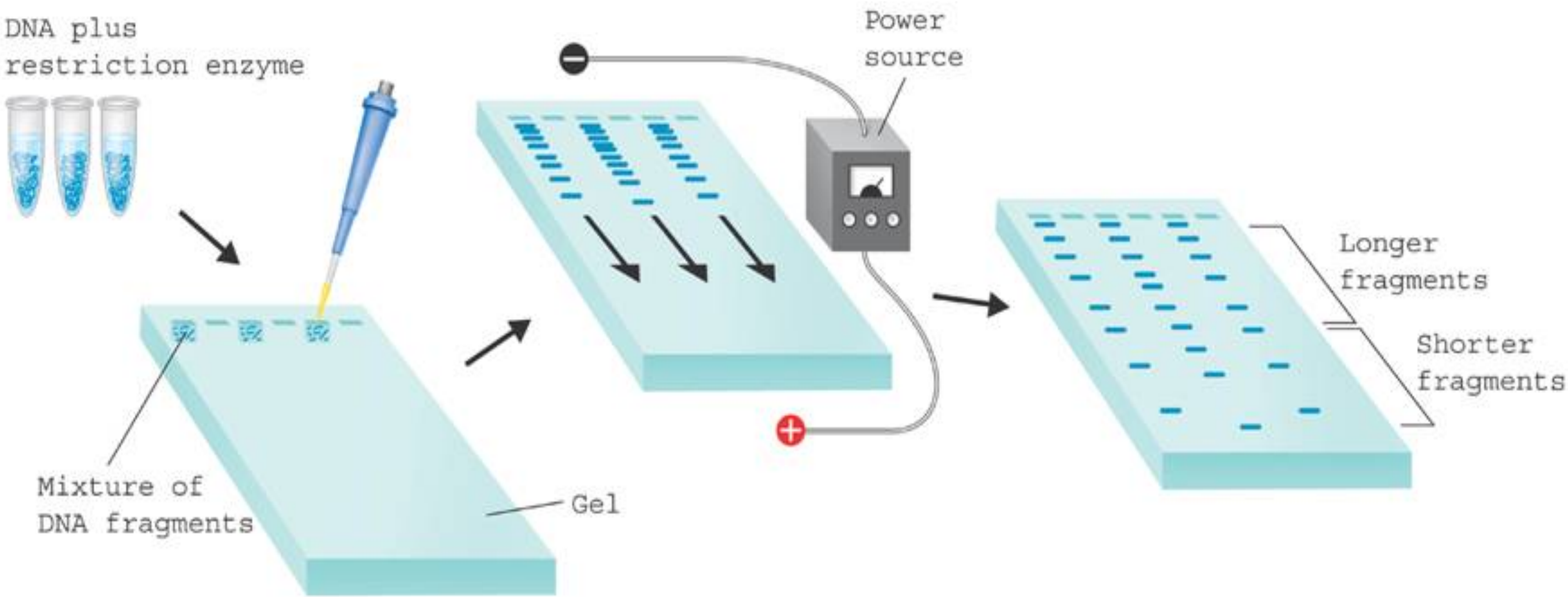
DNA Fingerprinting

- Once separated, a unique DNA pattern can be seen for every organism tested

- How might this be used to find the owner of a particular blood sample?



DNA fingerprinting



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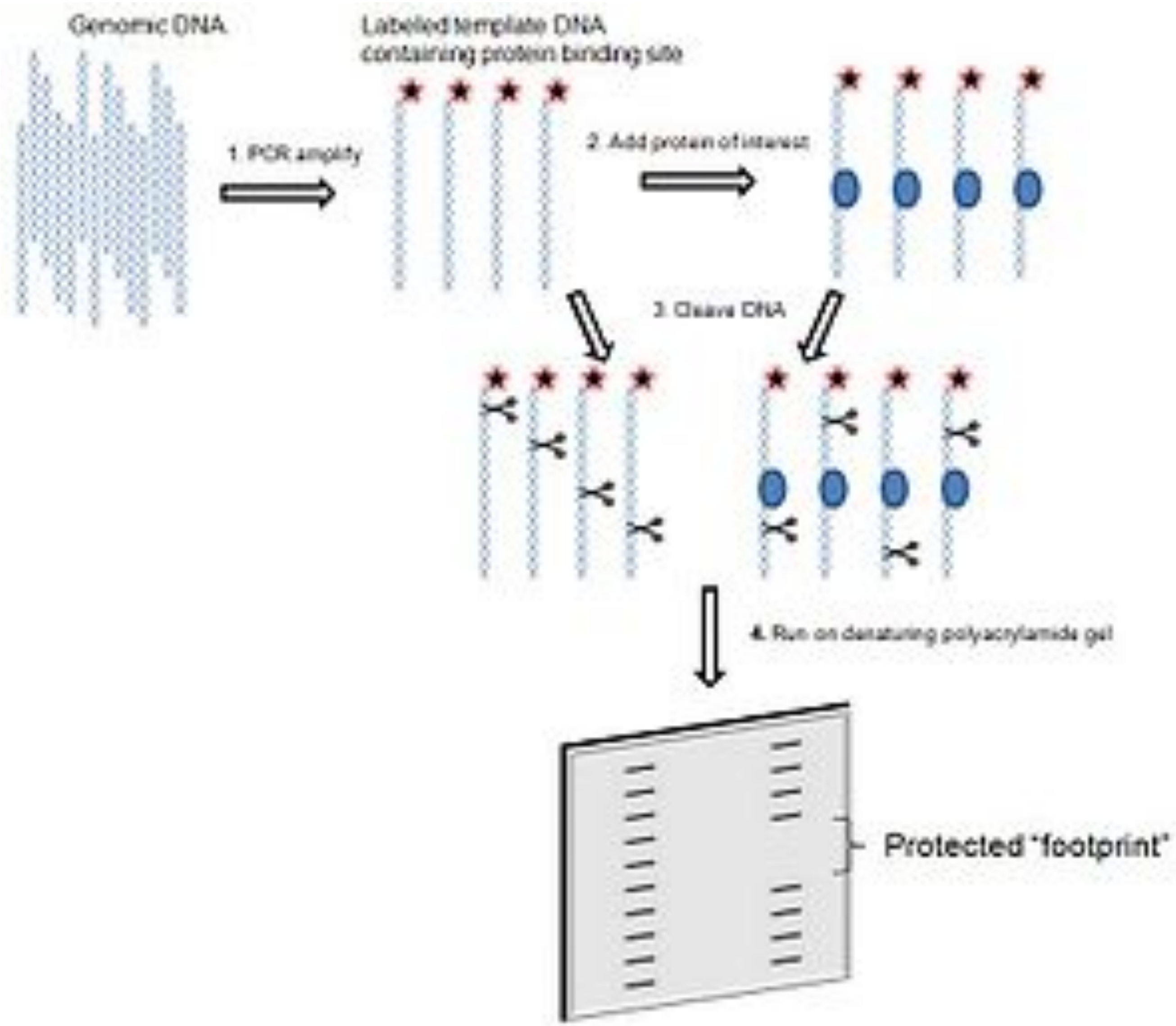
DNA footprinting

- A method of investigating the sequence specificity of DNA-binding proteins in vitro.
- Used to study protein-DNA interactions both outside and within cells.
- Help elucidate which proteins bind to promoters, enhancers, or silencers regions of DNA and unravel the complexities of transcriptional control.

DNA footprinting - Method

- To assess whether a given protein binds to a region of interest within a DNA molecule
  1. PCR amplify and label region of interest that contains a potential protein-binding site
     - ideally amplicon is between 50 to 200 base pairs in length.
  2. Add protein of interest to a portion of the labeled template DNA;
     - a portion should remain separate without protein, for later comparison
  3. Add a cleavage agent to both portions of DNA template.
     - The cleavage agent is a chemical or enzyme that will cut at random locations in a sequence independent manner.
     - The reaction should occur just long enough to cut each DNA molecule in only one location. A protein that specifically binds a region within the DNA template will protect the DNA it is bound to from the cleavage agent.
  4. Run both samples side by side on a PAGE.
     - The portion of DNA template without protein will be cut at random locations, and thus when it is run on a gel, will produce a ladder-like distribution.
     - The DNA template with the protein will result in ladder distribution with a break in it, the "footprint", where the DNA has been protected from the cleavage agent.

DNA footprinting - Method



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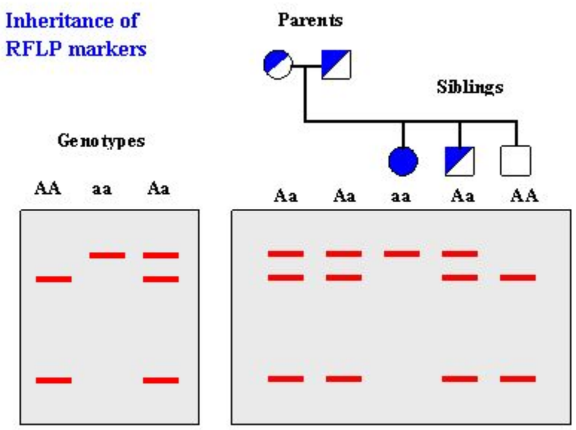
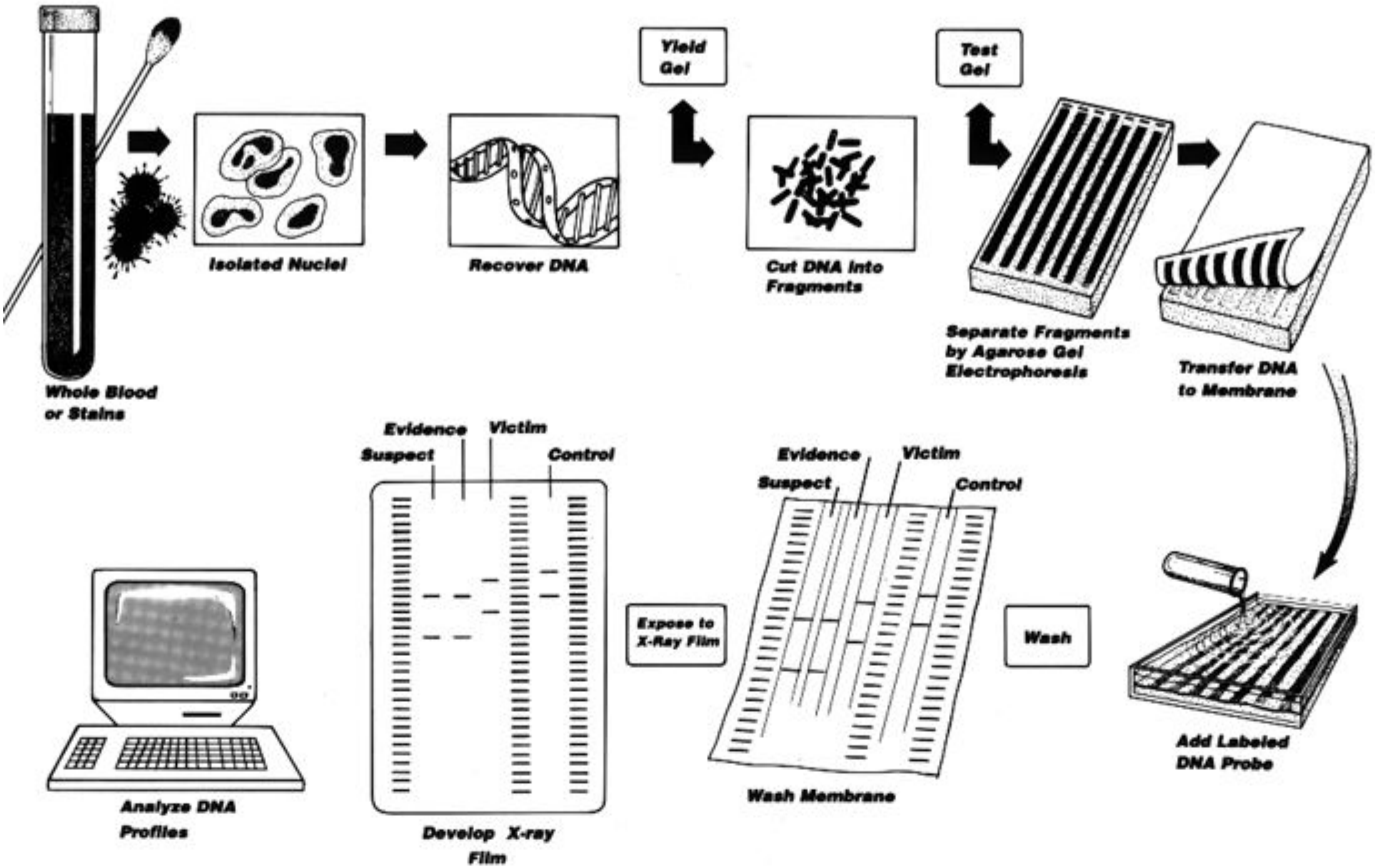
Restriction Fragment Length Polymorphism

What is RFLP?

- RFLP are differences in homologous DNA sequences detected by presence of fragments with varying lengths after digestion of DNA.
- RFLP is molecular marker specific to single restriction enzyme combination.
- In **RFLP analysis**, the DNA sample is broken into pieces (digested) by restriction enzymes and the resulting *restriction fragments* are separated according to their lengths by gel electrophoresis.
- RFLP analysis was an important tool in;
  - Genome mapping,
  - Localization of genes for genetic disorders,
  - Determination of risk for disease, and
  - Paternity testing`



RFLP - Method



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Karyotype

- The set of chromosomes of an individual.
- Chromosomes photographed during metaphase and arranged in a standard sequence

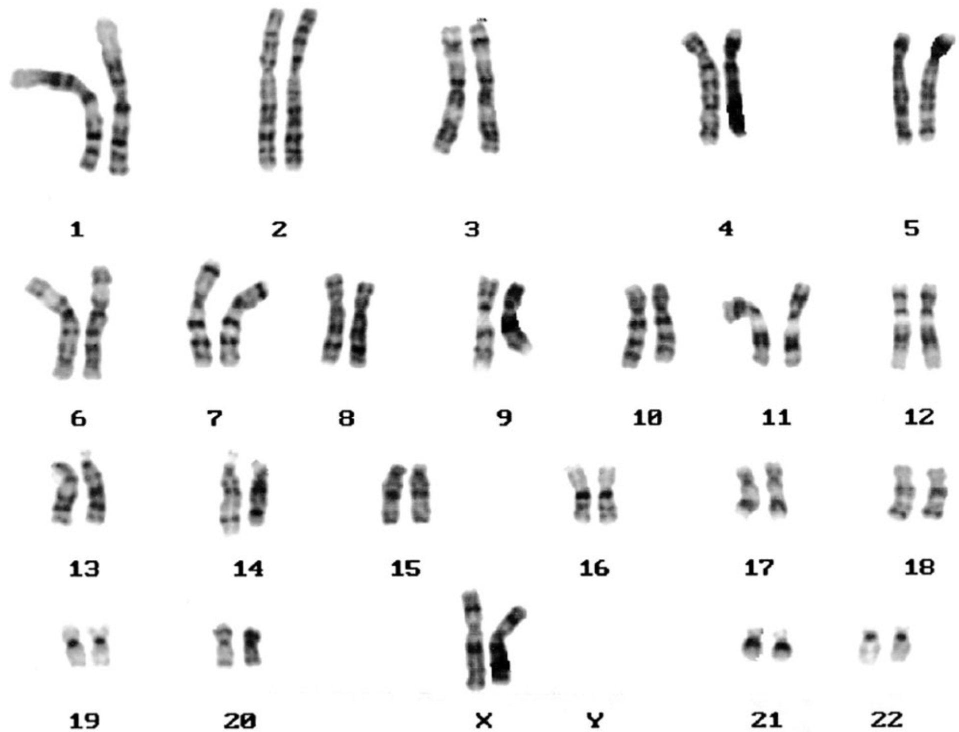
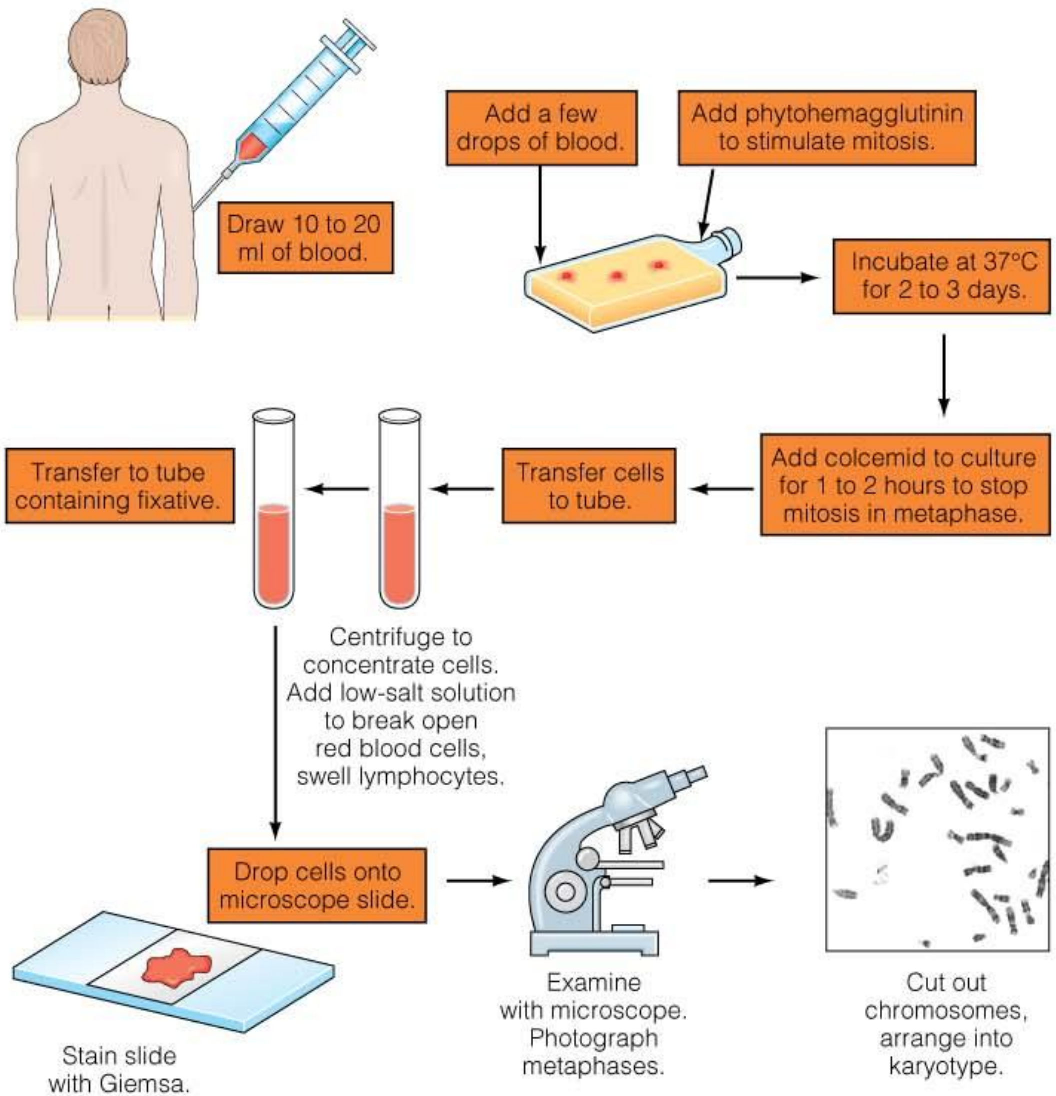


Fig. 6.3

Karyotyping: Creating a Karyotype



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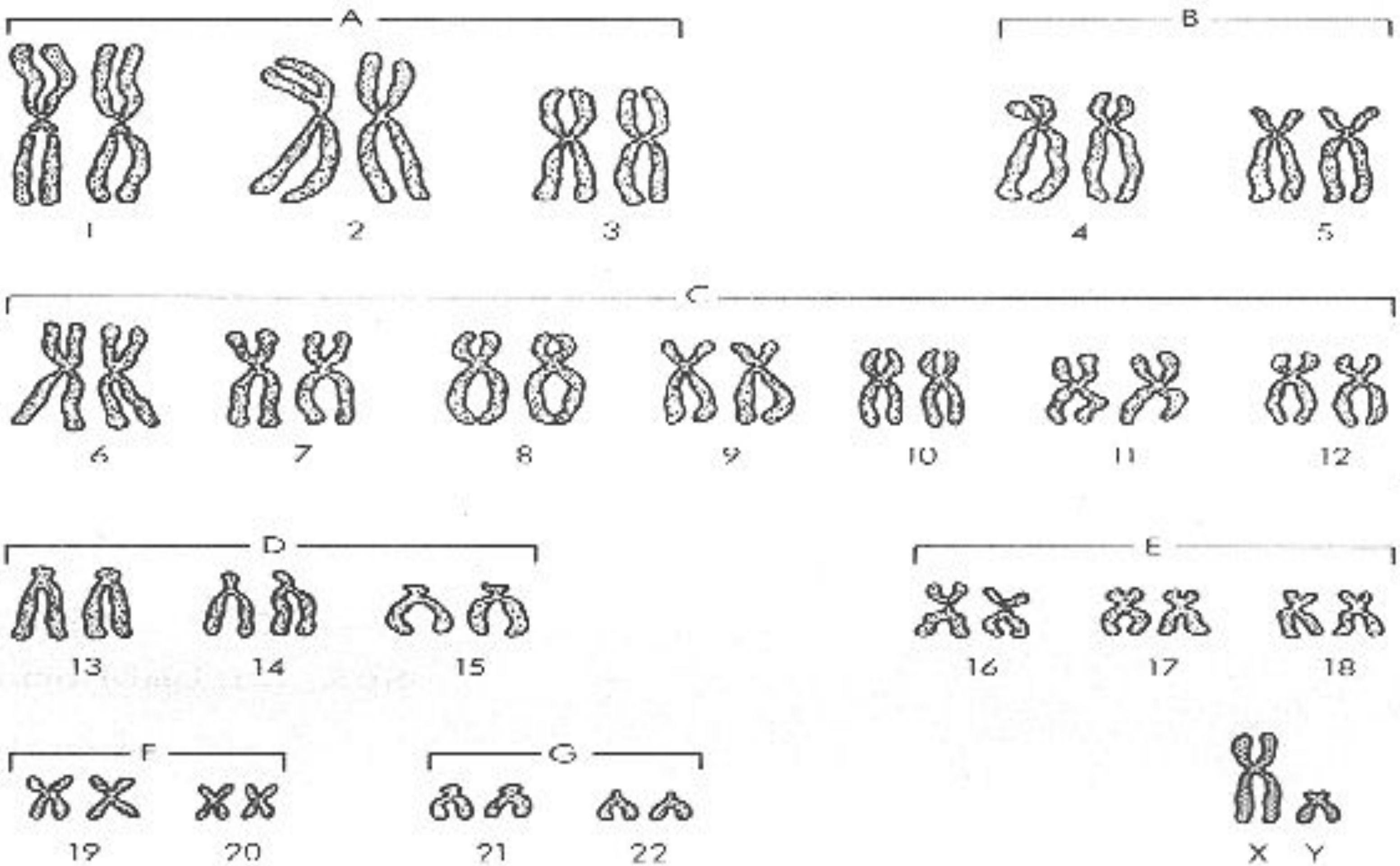
Karyotype of a normal male



Normal Male

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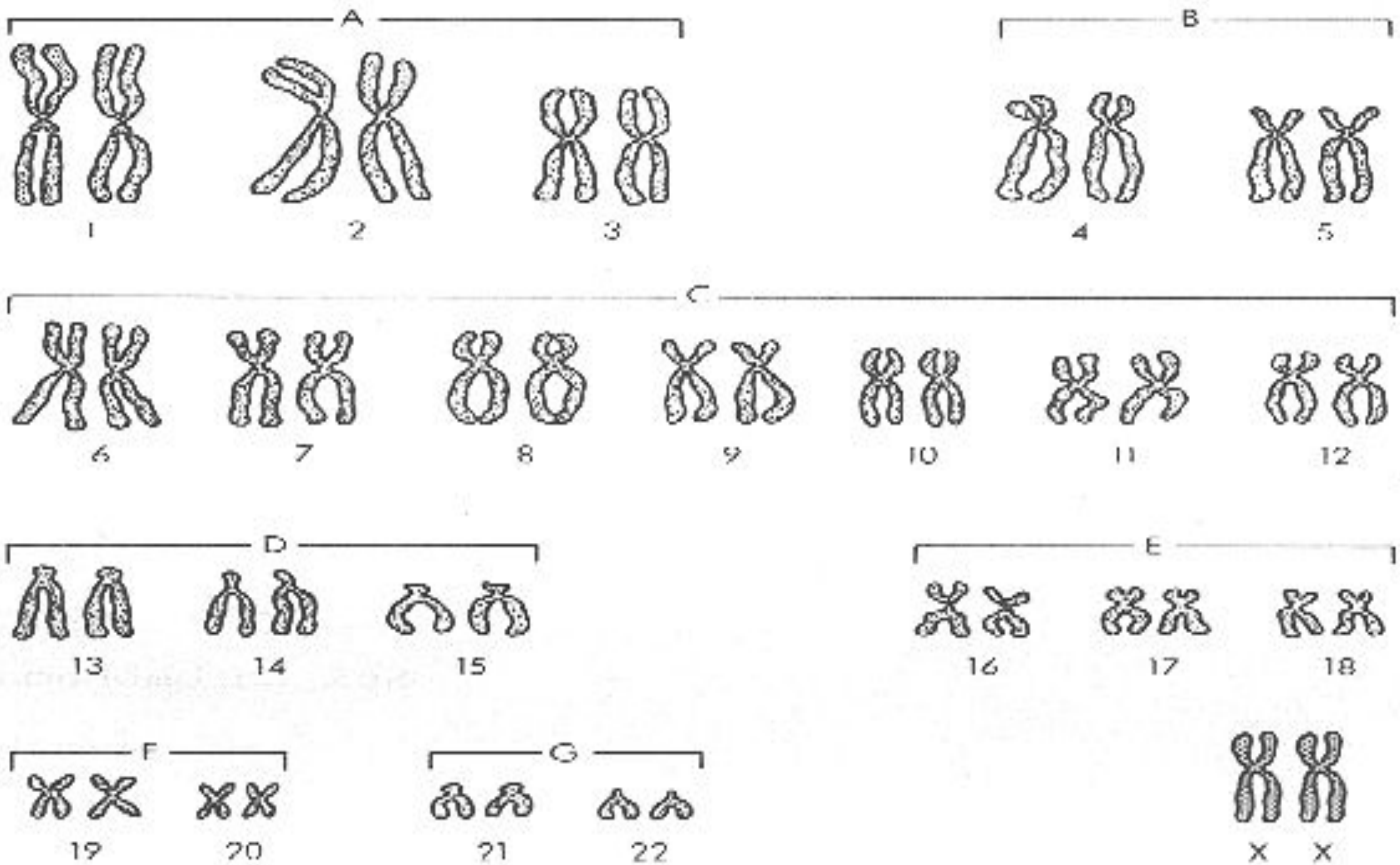
Karyotype of a normal female



Normal Female

Thank you…………

…………..Read More!